Universal Plastic Shredder V2.0

Electrical Control & Power System

Portland State University

Maseeh College of Engineering & Computer Science

Department of Electrical & Computer Engineering

Final Report — Rough Draft

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[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]

Contents

1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost $10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand deadman control, hardwired E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: $3,000 (EE portion approximately $1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., "shredder validated to X HP load with Y throughput rate") once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.

- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.
- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within $3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.

- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** bench/test unit is a GE 5KE182BC205B (3 HP, 230 V low-voltage connection, 1750 RPM, FLA 7.6 A); its nameplate drives the VFD motor parameters. [**TODO: Final production motor selection is TBD (3–5 HP per requirements).**]
- **Brake resistor:** 75 Ω, 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop and two-hand deadman start; motor contactor energized through the safety chain. Access guarding at the feed is handled by the ME team's mechanical design, so no separate electrical lid interlock is used.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset) → PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU –V) → VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) → PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI ↔ PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).

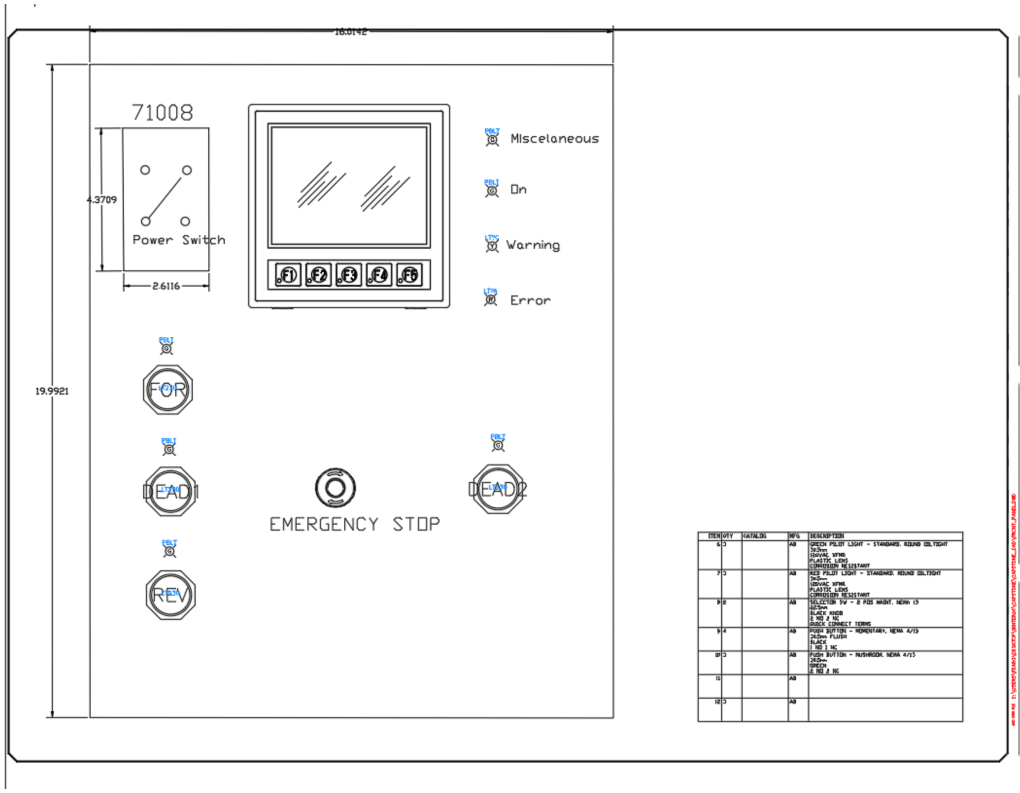

Figure 1: Operator control panel layout: HMI, power switch, forward/reverse and deadman pushbuttons, emergency stop, and status indicators.

[TODO: Add a system block diagram (power + signal flow) to complement the panel layout above.]

5 Hardware Design

5.1 Power Distribution

- Main disconnect / power switch: PowerTec 71008 two-pole (DPST) START/STOP switch, 120–230 VAC single-phase, rated 20 A / 3 HP max; switches both the line and neutral conductors.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90 °C ampacity columns of NEC Table 310.16 – well above the motor's 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG,

landed directly as straight stripped conductors at the terminal blocks (no ferrules used in this build).

- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

The Huanyang HY02D211B-T was selected over alternatives via a decision matrix (price, documentation, reliability). A brake resistor on the P/Pr terminals absorbs the regenerative energy from rapid reversal during the jam-clear cycle. The drive is programmed as shown in Table 1.

Table 1: Programmed VFD parameters (Huanyang HY02D211B-T).

Param	What it does	Value	Why this value
`PD141`	Motor rated voltage	230 V	Matches the GE test-motor nameplate (230 V low-voltage connection); sets the base of the V/Hz curve.
`PD142`	Motor rated current (FLA)	7.6 A	Nameplate full-load amps; sets the drive's current limit and electronic-overload reference.
`PD143`	Number of motor poles	4	Nameplate; 4-pole = 1800 RPM synchronous, so the drive resolves speed and slip correctly.
`PD144`	Motor rated speed	1750 RPM	Nameplate full-load speed; with the pole count it lets the drive estimate slip.

continued on next page . . .

Param	What it does	Value	Why this value
`PD002`	Run / frequency source	1 (external)	Speed comes from the external 0–10 V analog input (VI) driven by the PLC, not the keypad.
`PD052`	Relay output (FA/FC) function	02 (fault)	Closes the FA–FC relay on any fault so the PLC reads a jam/trip on input X5.
`PD054`	Analog output (VO) function	1 (current)	Outputs 0–10 V proportional to motor current to the PLC analog input for monitoring.
`PD123`	Over-torque action	3 (trip)	Over-torque self-trips the drive (which fires the fault relay); see the note below.
`PD124`	Over-torque threshold	150%	Set between no-load current (∼0.5 A) and measured jam current (>11 A) so normal shredding does not trip but a jam does.
`PD125`	Over-torque detection time	3.0 s	The over-torque must persist 3 s before tripping, so brief load spikes (feeding a piece) do not nuisance-trip.

[NOTE: The over-torque relay function (`PD052 = 12`) did not fire the relay in our specific Huanyang revision when `PD123 = 2` (continue running). Documented as a firmware-specific limitation (see Table 1). Working configuration uses `PD052 = 02` (fault indication) and `PD123 = 3` (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD's latched fault state between retry attempts.]

5.3 Safety Circuit

- E-stop: NC contact (E22B1) hardwired in series with the SD-N35 contactor coil. Pressing it drops the coil and opens the contactor, removing motor power independently of the PLC.

- Two-hand deadman: both buttons must be held continuously to enable the motor (evaluated by the PLC; see the Safety Functions assessment below).
- Access guarding: handled by the ME team's mechanical design (feed geometry), so no separate electrical lid interlock is used (this was a may-have, not a requirement).
- Motor contactor: the SD-N35 is energized through the hardwired safety chain, and the VFD is powered through it, so the safety chain has hardware precedence over the PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (`PD052 = 02`). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA–FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads `[ X5 ]` on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the independent hard-wired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

6 Software Design

6.1 PLC Ladder Logic Architecture

The control program runs on the H2-DM1E CPU in the DirectLOGIC 205 D2-09B-1 rack and was written in Do-more Designer. The program is 31 rungs long and is exported in `electrical/Circuit_logic.pdf`. It implements a state machine for normal operation plus a 3-strike jam recovery sequence and a hard-lockout path.

6.1.1 Variable and Coil Reference

The full variable map exported from the Do-more program is reproduced below. Discrete inputs (X), discrete outputs (Y), analog input/output words (WX/WY), internal coils (C), timers (T), counters (CT), retentive registers (V), and Do-more system bits (ST) are all used.

Address	Name in Ladder	Type	Description
X0	FORWARDS	DI	Forward pushbutton (green S2PR-P3G). HIGH = pressed.
X1	REVERSE	DI	Reverse pushbutton (yellow S2PR-P3Y). HIGH = pressed.
X2	DEADMAN_A_OK	DI	Deadman button A (blue #1). Must be held for DEADMEN_OK.
X3	DEADMAN_B_OK	DI	Deadman button B (blue #2). Must be held simultaneously with X2.
X4	STOP_PB	DI	E-stop NC contact monitor. LOW when E-stop is pressed. Used as NOT X4 in rungs 17, 18.
X5	OVERCURRENT	DI	VFD fault relay input (FA/FC, PD052=02). HIGH = VFD tripped. Triggers jam recovery sequence.
X7	CAP_TOUCH_TRIP	DI	Capacitive touch safety sensor (on-hand, wired). HIGH = safety zone breached. Immediately sets STOP_MOTOR, CTSI_TRIG.

Y0	FWD_LED	DO	Forward indicator LED (green SA-LDG). ON when FORWARDS_SET (C8) is HIGH.
Y1	REVERSE_LED	DO	Reverse indicator LED (yellow SA-LDY). ON when REVERSE_SET (C9) is HIGH.
Y2	DMAN_A_LED	DO	Deadman A LED (blue #1). ON when (X2 OR C23) AND NOT Y4.
Y3	DMAN_B_LED	DO	Deadman B LED (blue #2). ON when (X3 OR C24) AND NOT Y4.
Y4	E_LIGHT	DO	Error / fault indicator (Wamco 525 panel LED). SET by E-stop open, CAP_TOUCH, and CT0.Done. RST by ERROR_RESET (C18).
Y5	VFD_RESET	DO	VFD fault reset output (RST terminal). Energized while DEADMEN_OK AND X5 are both HIGH (rung 19).
Y6	VFD_RUN_FWD	DO	VFD forward run (FOR terminal). Energized by rung 9 when all permissives are met and no fault/pause active.
Y7	VFD_RUN_RVS	DO	VFD reverse run (REV terminal). Energized by rung 10 via REVERSE_SET or unconditionally by UNJAM_R (C13).
WX0	Analog input word	AI word	Raw ADC from F2-08AD-1 CH1 (VFD VO, motor current). Input to rung 3 SCALE (0–4095 → 0–140).

WY0	Analog output word	AO word	Raw DAC to F2-08DA-2 CH1 (VFD VI, speed reference). Output of rung 2 SCALE (V2000 0–1600 → 0–4095).
C0	DEADMEN_OK	BOOL	HIGH when (X2 OR C23) AND (X3 OR C24). Both deadman paths held.
C1	HMI_STP_REQ	BOOL	HMI STOP button request. Rung 7: NOT C1 gates RUN_REQUEST_SET.
C3	PERMISSIVE_OK	BOOL	Mirrors DEADMEN_OK (C0). Used as interlock in rungs 7 and 8.
C8	FORWARDS_SET	BOOL latch	Forward direction selected. SET by X0 (rung 13) or on first scan init (rung 4). RST by X1.
C9	REVERSE_SET	BOOL latch	Reverse direction selected. SET by X1 (rung 14). RST by X0. RSTR clears C9–C30 on first scan.
C10	JAM_LATCH	BOOL	Output of T0 (OFFDTMR). HIGH while overcurrent is active; stays HIGH 2 s after X5 drops. Blocks Y6, drives T1.
C11	init flag	BOOL	SET on first scan by $FirstScan (rung 1 RST, rung 4 triggers init). Used to initialize FORWARDS_SET and clear C9–C30.

C12	RUN_REQUEST	BOOL latch	Motor run command. SET by rung 7, RST by rung 8 (NOT X4 or NOT PERMISSIVE_OK). Also shown as MOTOR_STATUS on HMI.
C13	UNJAM_R	BOOL	Output of T1 (OFFDTMR, 2 s). HIGH while JAM_LATCH is HIGH; stays HIGH 2 s after JAM_LATCH drops. Energizes Y7 (reverse) unconditionally.
C15	STOP_MOTOR	BOOL latch	Hard stop flag. SET by CAP_TOUCH or CT0.Done (lockout). RST by ERROR_RESET (C18). Blocks Y6 and Y7 in rungs 9, 10.
C16	LOCKOUT_TMR_STRT	BOOL latch	Enables T2 accumulating timer. SET when UNJAM_R active and T2 not timing. RST when T2.DN fires (30 s accumulated).
C17	PAUSE_MOTOR	BOOL	Output of T3 (OFFDTMR, 2 s). Triggered by T1.DN rising edge. Blocks Y6 for 2 s after reverse completes (coast-to-stop buffer).
C18	ERROR_RESET	BOOL	Fault reset coil (from HMI ERROR_OFF button, Screen 3). RSTs STOP_MOTOR, E_LIGHT, JAMMED_MOTOR, CTSI_TRIG, CHANGE_SCREEN, CT0.

C20	JAMMED_MOTOR	BOOL latch	Lockout flag. SET when CT0.Done fires (3rd jam). Shown on Screen 3 SHREDDER JAM indicator. RST by ERROR_RESET.
C21	CTSI_TRIG	BOOL latch	Capacitive touch triggered. SET by X7 (rung 28) or ERROR_RESET path. Shown on Screen 3. RST by ERROR_RESET.
C22	CHANGE_SCREEN	BOOL latch	Triggers HMI screen change to error screen (Screen 3). Rung 29: sets V100=3 and starts T4 (3 s). T4.DN resets V100 to 0.
C23	TESTMAN_A_OK	BOOL	HMI soft deadman A (from Test Screen PUSHBUTTON A toggle, D-MAN A tag). ORed with X2 in rung 5.
C24	TESTMAN_B_OK	BOOL	HMI soft deadman B (from Test Screen PUSHBUTTON B toggle, D-MAN B tag). ORed with X3 in rung 5.
C25	OVERCURRENT_TEST	BOOL	HMI soft overcurrent trigger (Test Screen OVERCURRENT momentary button). ORed with X5 in rung 19.
C26	E_STOP_TRIG	BOOL	E-stop triggered flag. SET by rung 17 when NOT X4. Shown on Screen 3 EMERGENCY STOP indicator.
C99	T4 output	BOOL	Output of T4 off-delay timer (3 s). Active 3 s after CHANGE_SCREEN sets. Not used further in current build.

T0	Overcurrent off-delay	OFFDTMR 2 s	Input: DEADMEN_OK AND (X5 OR C25). Output: JAM_LATCH (C10). Extends jam latch 2 s after overcurrent clears.
T1	Unjam reverse off-delay	OFFDTMR 2 s	Input: JAM_LATCH (C10). Output: UNJAM_R (C13). Drives reverse for 2 s after JAM_LATCH drops.
T2	Lockout accumulating timer	TMRA 30 s	Accumulates total time LOCKOUT_TMR_STRT (C16) is active. T2.DN at 30 s resets C16, stops counting new jams.
T3	Post-reverse pause	OFFDTMR 2 s	Triggered by T1.DN rising edge. Output: PAUSE_MOTOR (C17). Holds Y6 LOW for 2 s after reverse finishes.
T4	Screen-change timeout	OFFDTMR 3 s	Input: CHANGE_SCREEN (C22). After 3 s, T4.DN fires rung 30: resets V100 to 0 (HMI returns from error screen).
CT0	JAM_DETECT	CTU preset 3	Counts jam events. UP input: JAM_LATCH AND NOT T2.DN. RST input: ERROR_RESET (C18). CT0.Done (N0=3) triggers lockout.
V100	ERROR_SCREEN	Int (BCD)	HMI screen selector written by ladder. Rung 29 writes 3 (→ Screen 3), rung 30 writes 0 after T4 (3 s).

V2000	RPM_CTRL	Int 16	Speed setpoint (0–1600 counts). Source for rung 2 SCALE → WY0. Written by HMI IncDec buttons.
V3000	CURRENT	Int 16	Scaled motor current (0–140). Output of rung 3 SCALE from WX0. Displayed on Screen 2 CURRENT field.
ST0	$FirstScan	System bit	Do-more system coil HIGH for exactly one scan on power-up. Used in rung 1 to RST C11 (init trigger).
ST1	$On	System bit	Do-more system coil always HIGH. Used in rungs 2 and 3 to run SCALE instructions every scan.

6.1.2 State Machine

- **IDLE** → RUNNING when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → JAM_DETECTED on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → RST_PULSE if jam count < 3, → LOCKOUT if jam count ≥ 3.
- **RST_PULSE** (500 ms) → REVERSE (2 s).
- **REVERSE** → RESUME (brief delay) → back to RUNNING.
- **LOCKOUT** → IDLE only via dedicated reset button.

6.1.3 Jam Counter Reset Logic

- Counter (CT0) increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets when motor runs cleanly for 30 continuous seconds (T2 accumulating timer).

- Also resets via the HMI Screen 3 ERROR_OFF button (ERROR_RESET, C18).
- Whichever condition fires first wins.

6.1.4 Startup Mask

- For first scan after PLC power-up, the FORWARDS_SET / REVERSE_SET pair is initialized via rung 4 (defaults to forward direction).
- In the active-high NO X5 wiring, no boot-time mask is required because the relay is naturally open at startup.

6.1.5 Rung-by-Rung Description with Ladder Logic Export

The 31 rungs are reproduced below directly from the Do-more Designer export (`electrical/Circuit_logic` with the actual ladder image on each page paired with a description of the rungs it contains. This lets a reviewer see the rung graphic next to the variable map (above) and the narrative below.

Page 1 of 7 — Rungs 1–5: Power-up Init and Deadman Permissive

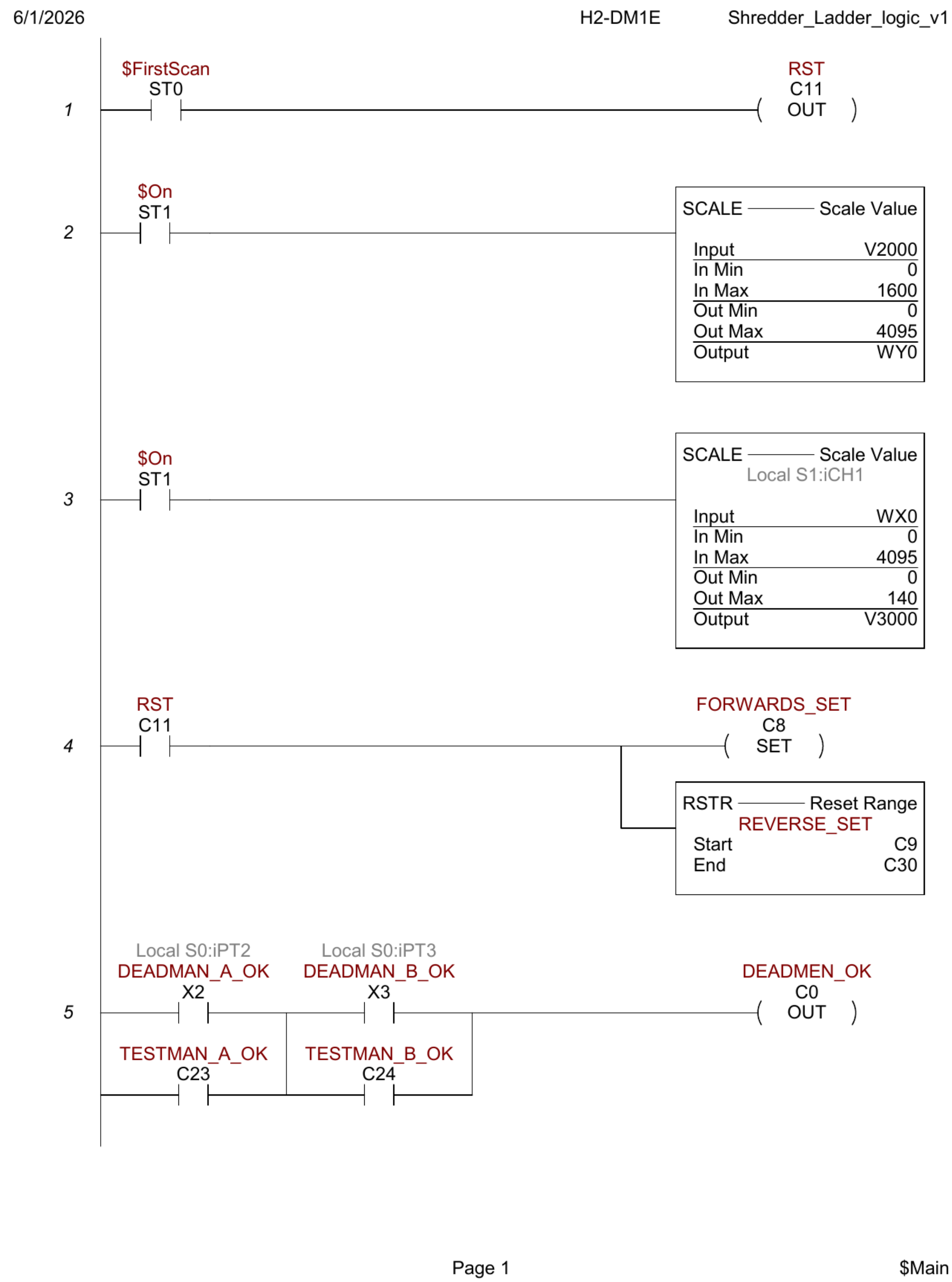

Figure 2: Ladder export page 1. Power-up init and deadman permissive (rungs 1–5).

- **Rung 1** ($FirstScan → RST C11): on the very first scan after power-up, C11 is reset, triggering the init path in rung 4.
- **Rung 2** ($On → SCALE V2000 [0–1600] → WY0 [0–4095]): every scan, the HMI speed setpoint V2000 is linearly scaled to raw DAC counts for the F2-08DA-2 CH1 → VFD VI.
- **Rung 3** ($On → SCALE WX0 [0–4095] → V3000 [0–140]): raw ADC from F2-08AD-1 CH1 (VFD VO motor current) is scaled to a 0–140 display value stored in V3000 (CURRENT tag).
- **Rung 4** (NOT C11 → SET FORWARDS_SET + RSTR C9–C30): initializes FORWARDS_SET and clears C9–C30 on first power-cycle scan.
- **Rung 5** ((X2 OR C23) AND (X3 OR C24) → OUT C0): computes DEADMEN_OK by AND-ing the two deadman paths (physical OR HMI-soft test buttons).

Page 2 of 7 — Rungs 6–10: RUN_REQUEST Gating and Run Outputs

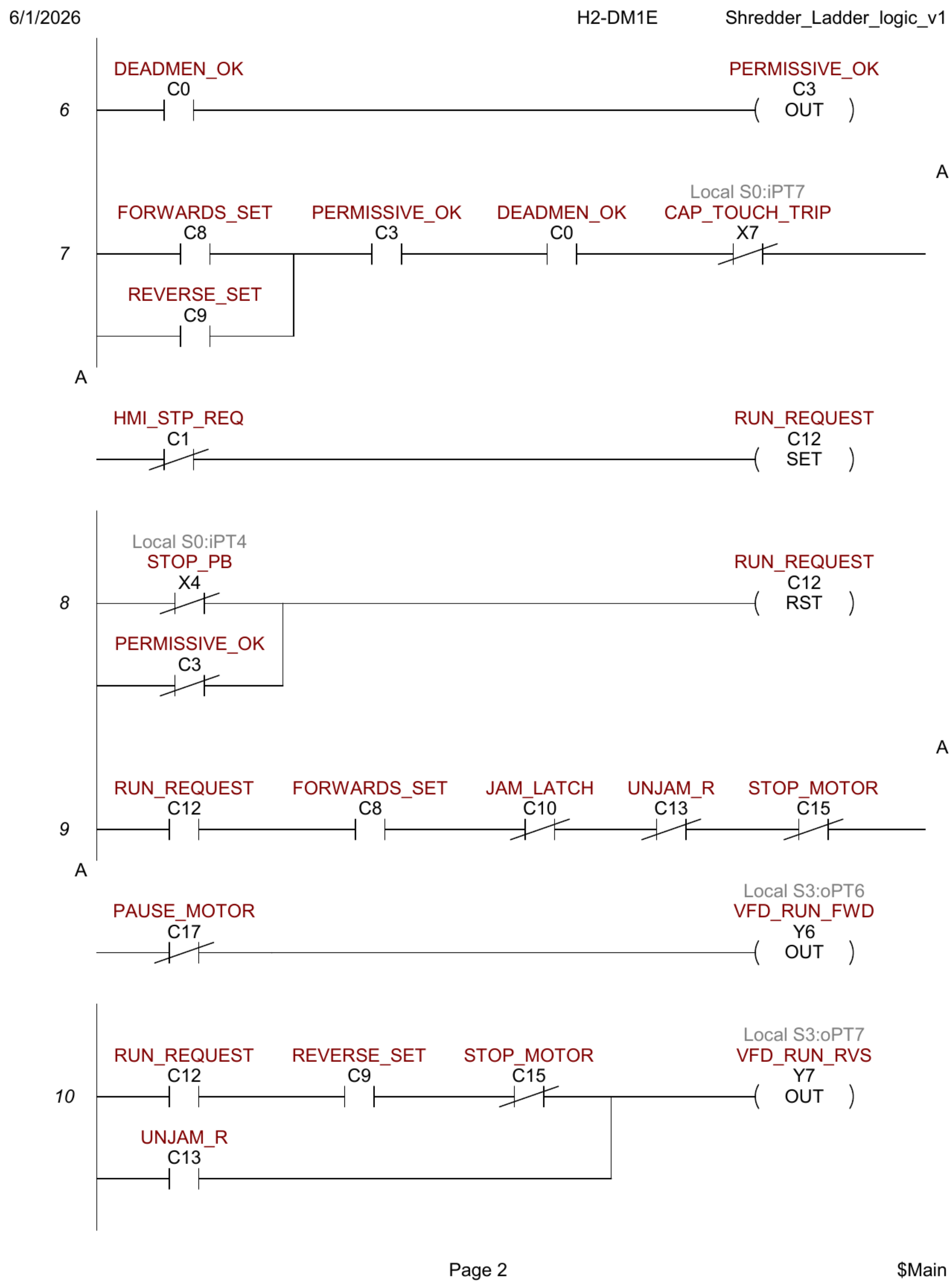

Figure 3: Ladder export page 2. RUN_REQUEST gating and run outputs Y6/Y7 (rungs 6–10).

- **Rung 6** (C0 → OUT C3): mirrors DEADMEN_OK to PERMISSIVE_OK so the permissive logic is referenced separately from the deadman coil.
- **Rung 7** ((C8 OR C9) AND C3 AND C0 AND NOT X7 → SET C12): RUN_REQUEST is set when a direction is selected, permissives are OK, and the capacitive touch is not tripped. The HMI STOP button (NOT C1) is a separate inhibit path.
- **Rung 8** (NOT X4 OR NOT C3 → RST C12): RUN_REQUEST is immediately reset if the E-stop opens (X4 LOW) or the deadman permissive drops.
- **Rung 9** (C12 AND C8 AND NOT C10 AND NOT C13 AND NOT C15 AND NOT C17 → OUT Y6): forward run output. Requires RUN_REQUEST + FORWARDS_SET + no active jam + not in reverse unjam stroke + no hard stop + post-reverse buffer elapsed.
- **Rung 10** ((C12 AND C9 AND NOT C15) OR C13 → OUT Y7): reverse run has two paths: manual reverse (RUN_REQUEST AND REVERSE_SET AND NOT STOP_MOTOR), or unconditional unjam reverse (UNJAM_R alone).

Page 3 of 7 — Rungs 11–16: Indicator LEDs and Direction Latches

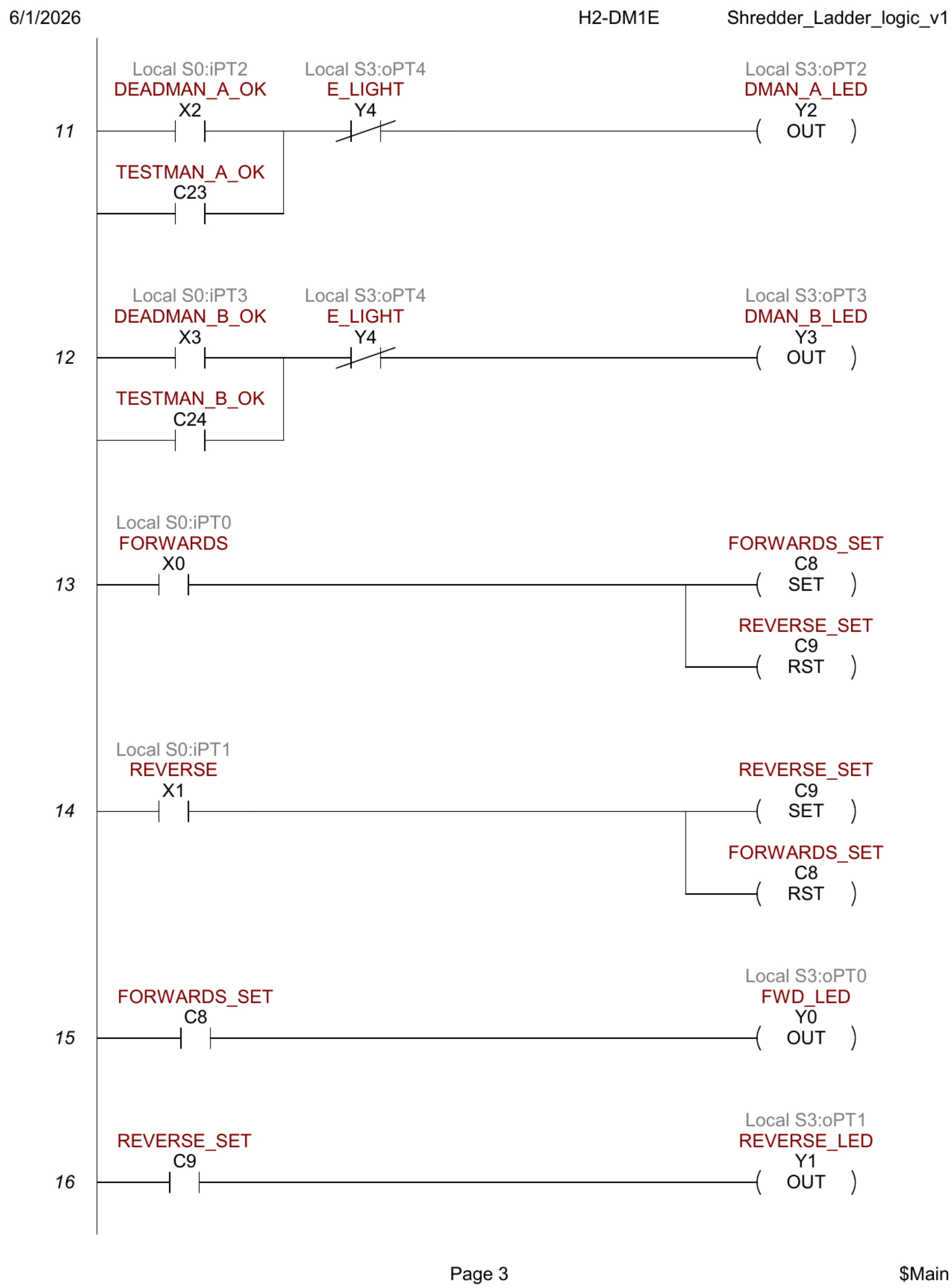

Figure 4: Ladder export page 3. Indicator LEDs and direction latches (rungs 11–16).

- **Rung 11** ((X2 OR C23) AND NOT Y4 → OUT Y2): deadman A LED lights when the button is pressed but only if the fault indicator (Y4) is not lit.
- **Rung 12** ((X3 OR C24) AND NOT Y4 → OUT Y3): identical pattern for deadman B.
- **Rung 13** (X0 → SET C8 + RST C9): forward pushbutton latches FORWARDS_SET and clears REVERSE_SET (mutual exclusion).
- **Rung 14** (X1 → SET C9 + RST C8): reverse pushbutton is the mirror image.
- **Rung 15** (C8 → OUT Y0): forward indicator LED ON whenever FORWARDS_SET is latched.
- **Rung 16** (C9 → OUT Y1): reverse indicator LED ON whenever REVERSE_SET is latched.

Page 4 of 7 — Rungs 17–21: E-stop Response, Jam Detection, and Unjam Stroke

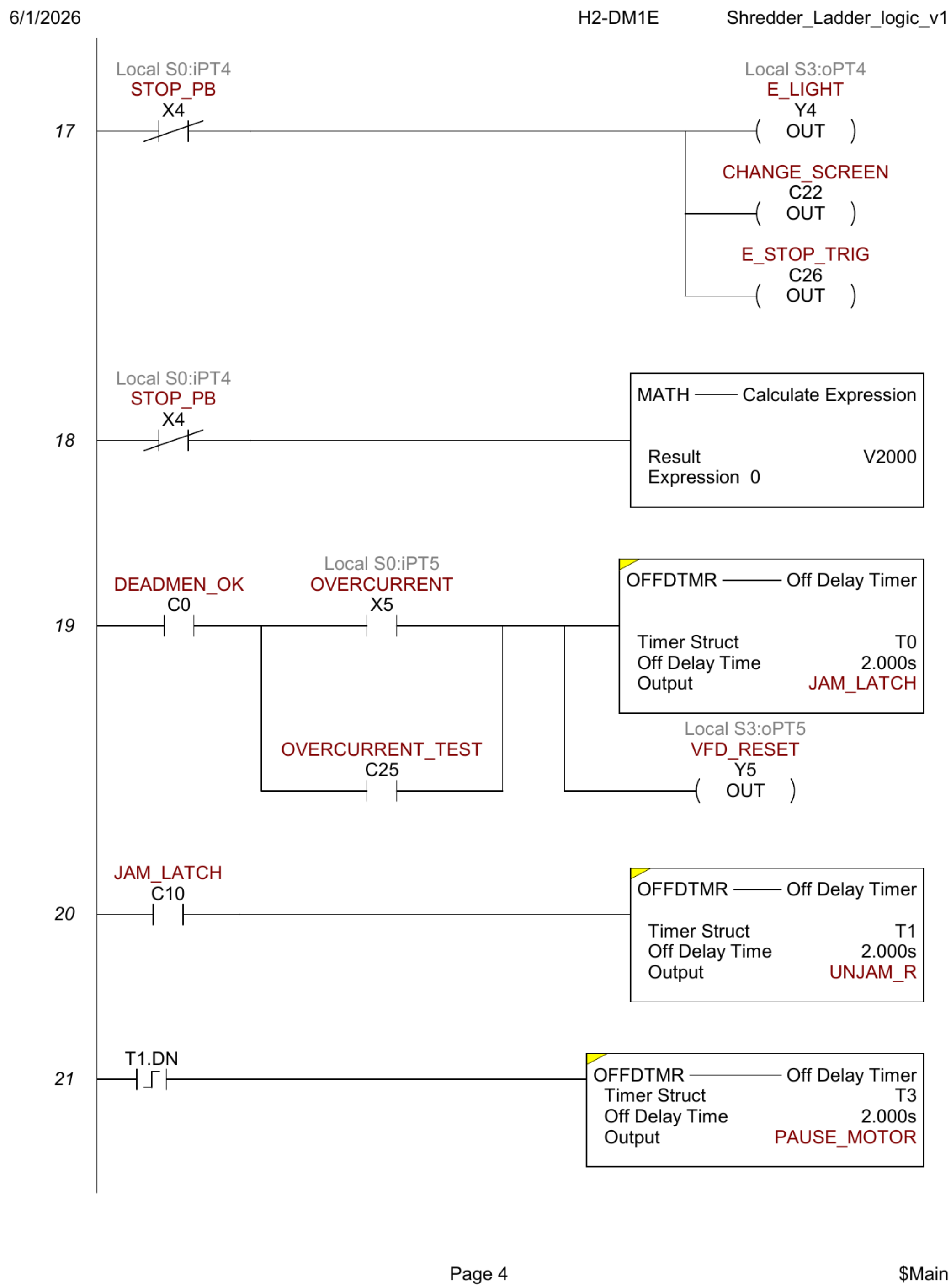

Figure 5: Ladder export page 4. E-stop response, jam detection, and unjam stroke (rungs 17–21).

- **Rung 17** (NOT X4 → OUT Y4 + OUT C22 + OUT C26): when E-stop opens (X4 LOW), immediately light the fault LED, set the screen-change flag for HMI Screen 3, and set the E_STOP_TRIG indicator.
- **Rung 18** (NOT X4 → MATH V2000=0): when E-stop opens, zero the speed setpoint to prevent a speed step on power restore.
- **Rung 19** (C0 AND (X5 OR C25) → OFFDTMR T0 (2 s) → C10 + OUT Y5): jam detection. While deadman is held and the VFD fault relay is active (or HMI soft test is active), JAM_LATCH (C10) goes HIGH immediately and stays HIGH 2 s after the input drops; Y5 (VFD_RESET) is energized directly to pulse the VFD RST terminal.
- **Rung 20** (C10 → OFFDTMR T1 (2 s) → C13): unjam reverse stroke timer. UNJAM_R (C13) stays HIGH for 2 s after JAM_LATCH drops, so rung 10 forces Y7 (reverse) for at least 2 s of recovery.
- **Rung 21** (T1.DN rising edge → OFFDTMR T3 (2 s) → C17): post-reverse pause. When UNJAM_R finishes, T3 holds PAUSE_MOTOR HIGH for 2 s so the motor can coast to rest before re-accelerating forward.

Page 5 of 7 — Rungs 22–26: Accumulating Lockout Timer and Jam Counter

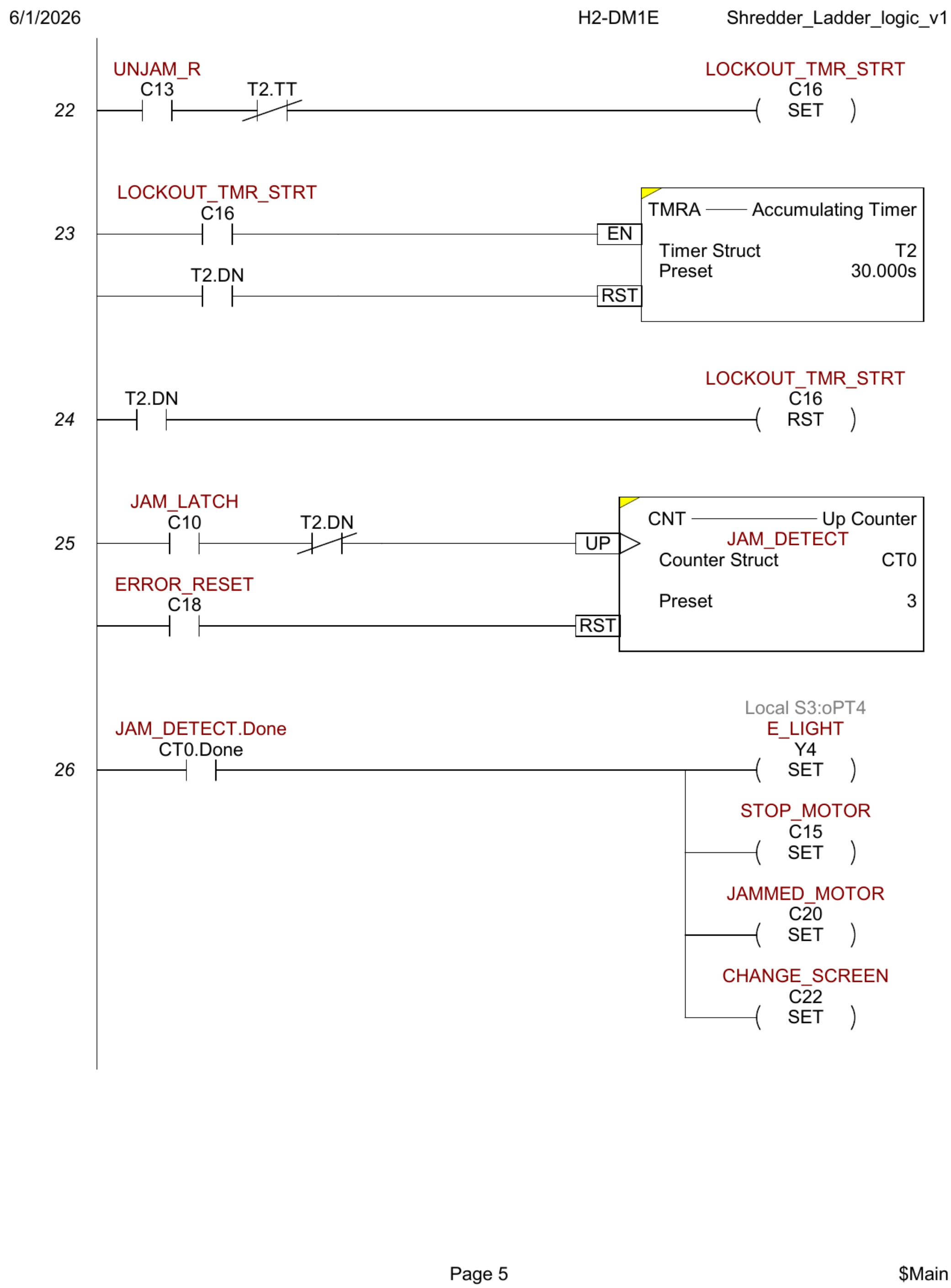

Figure 6: Ladder export page 5. Accumulating lockout timer and jam counter (rungs 22–26).

- **Rung 22** (C13 AND NOT T2.TT → SET C16): starts the jam-count accumulating timer. The NOT T2.TT condition prevents re-triggering if T2 is mid-count.
- **Rung 23** (C16 → TMRA T2 EN; T2.DN → RST T2): 30-second accumulating timer. T2.DN resets itself and rung 24 clears C16, stopping accumulation.
- **Rung 24** (T2.DN → RST C16): after 30 s of accumulated unjam activity, stop accumulation. New JAM_LATCH events no longer increment CT0.
- **Rung 25** (C10 AND NOT T2.DN → CNT UP CT0 (preset 3) [RST: C18]): jam counter increments each time JAM_LATCH goes HIGH while T2 has not completed. Preset 3 means CT0.Done fires after the third jam.
- **Rung 26** (CT0.Done → SET Y4 + SET C15 + SET C20 + SET C22): lockout on third jam. Y4 fault LED on permanently, STOP_MOTOR hard-stops the motor (rungs 9 and 10 block), JAMMED_MOTOR triggers the Screen 3 indicator, CHANGE_SCREEN switches to Screen 3.

Page 6 of 7 — Rungs 27–29: Fault Reset, Capacitive Touch, HMI Screen Change

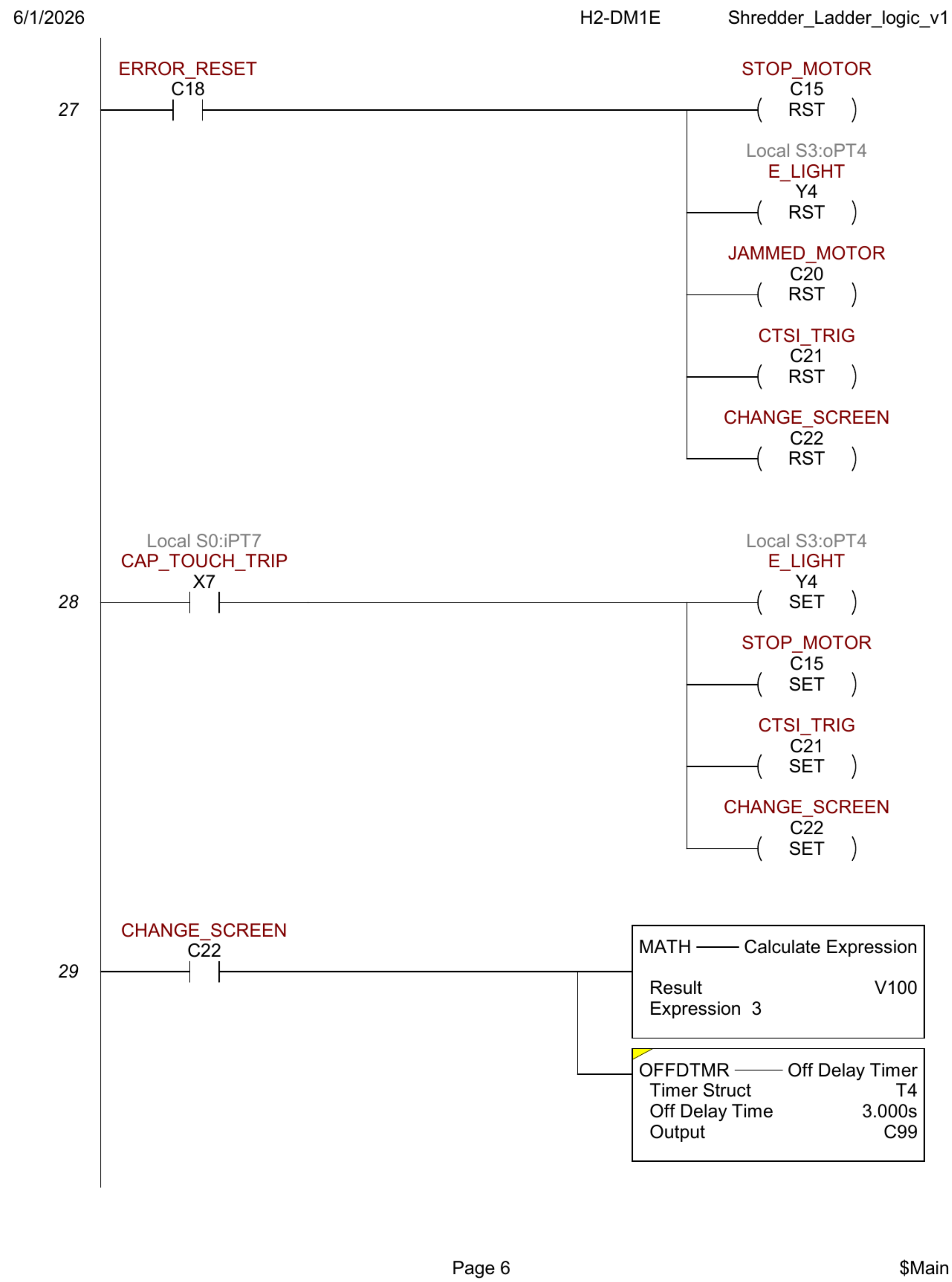

Figure 7: Ladder export page 6. Fault reset path, capacitive touch trip, and HMI screen change (rungs 27–29).

- **Rung 27** (C18 → RST C15 + RST Y4 + RST C20 + RST C21 + RST C22): fault reset path (from HMI Screen 3 SHREDDER JAM or CAP TOUCH button, tag ERROR_OFF = C18).
- **Rung 28** (X7 → SET Y4 + SET C15 + SET C21 + SET C22): capacitive touch safety trip. Any HIGH on X7 immediately sets all four lockout flags.
- **Rung 29** (C22 → MATH V100=3 + OFFDTMR T4 (3 s) → C99): HMI screen navigation. MATH writes 3 to V100 (ERROR_SCREEN tag) so the HMI switches to Screen 3.

Page 7 of 7 — Rungs 30–31: Screen Reset and End of Program

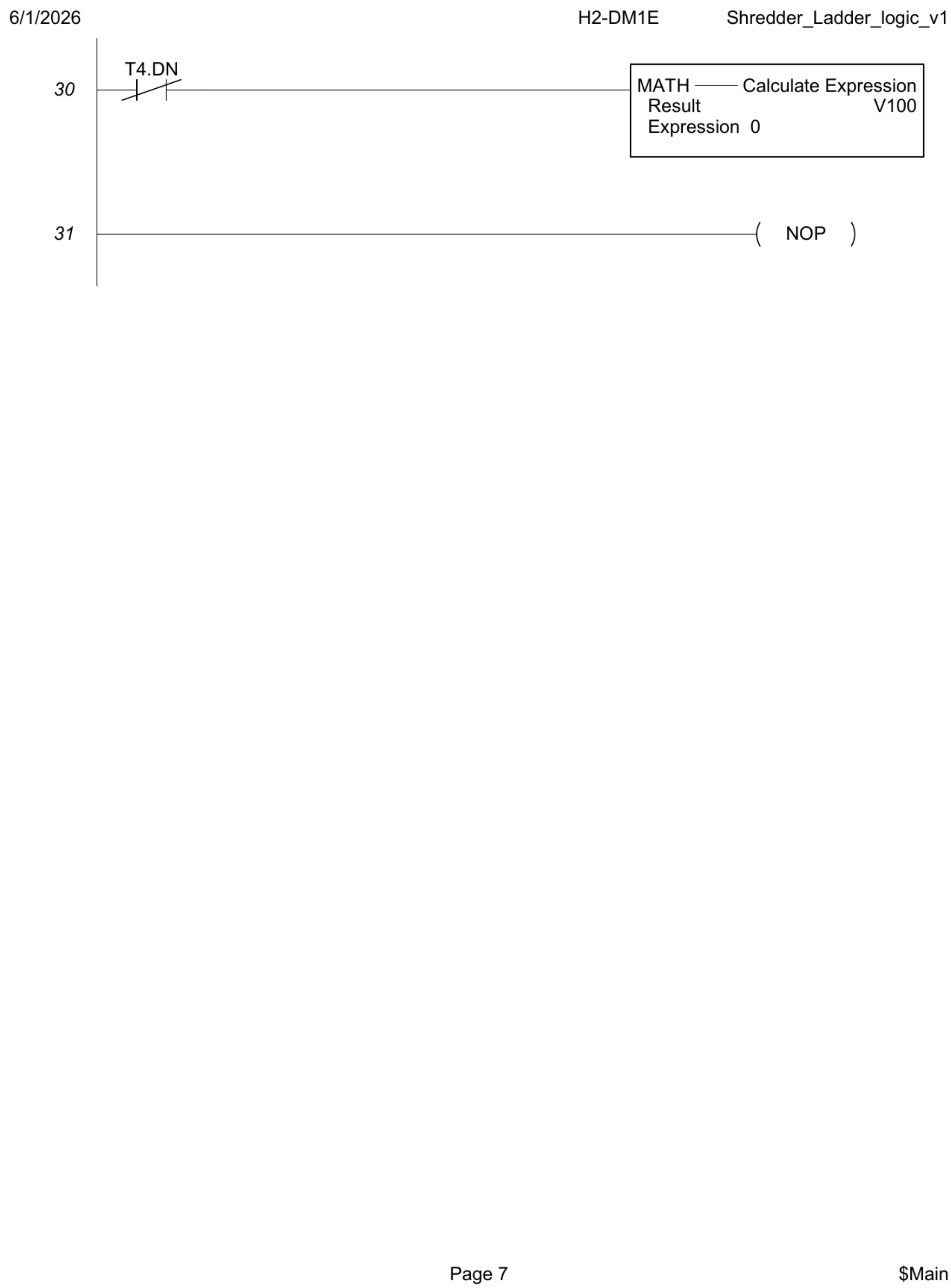

Figure 8: Ladder export page 7. Screen reset timeout and end-of-program NOP (rungs 30–31).

- **Rung 30** (T4.DN → MATH V100=0): after 3 s, V100 is reset to 0, releasing the HMI from the forced screen-3 display.
- **Rung 31** (NOP): no-operation placeholder marking the end of the program.

6.1.6 Jam Recovery Sequence — Complete Timeline

The trace below shows exactly what happens from overcurrent detection through resume, based on the actual timer values and rung logic:

Time	Event / Signal	Active outputs and coils
$t = 0$	X5 (VFD fault relay) goes HIGH while C0 (DEADMEN_OK) is HIGH	T0 input HIGH → C10 (JAM_LATCH) HIGH; Y5 (VFD_RESET) HIGH; C13 (UNJAM_R) HIGH; Y7 (reverse) HIGH; Y6 LOW
$t = 0^+$	VFD fault cleared by Y5 pulse; X5 drops LOW	T0 off-delay starts (2 s); Y5 goes LOW; C10 stays HIGH; C13 stays HIGH; Y7 stays HIGH
$t = 2$ s	T0 off-delay expires → C10 (JAM_LATCH) goes LOW	C10 LOW → T1 off-delay starts (2 s); C13 (UNJAM_R) stays HIGH; Y7 stays HIGH; CT0 increments
$t = 4$ s	T1 off-delay expires → C13 (UNJAM_R) goes LOW; T1.DN rises	Y7 goes LOW (reverse stops); T1.DN rising edge triggers T3; C17 (PAUSE_MOTOR) HIGH
$t = 6$ s	T3 off-delay expires → C17 (PAUSE_MOTOR) goes LOW	Rung 9 NOT C17 condition clears; Y6 (forward) re-energizes if deadman still held

Lockout at $t = 0$ of 3rd jam	CT0.ACC = 3 → CT0.Done HIGH	C15 (STOP_MOTOR) SET; C20 (JAMMED_MOTOR) SET; Y4 SET; C22 → Screen 3. Motor locked until C18 (ERROR_RESET) from HMI.

6.2 HMI Programming

The HMI is an AutomationDirect EA1-T4CL 4-inch C-more Micro-Graphic touch panel running the `HMI_413.mgp` project, connected to the H2-DM1E CPU over the Do-more serial protocol. The project has 4 screens and 27 PLC tags.

6.2.1 HMI Tag Database

Every PLC address visible on the HMI is documented below. Tags without a PLC address are local HMI variables (used for navigation, transient state, or labels).

#	Tag Name	Data Type	PLC Address	What it represents
1	AUTO	Discrete	—	HMI-local; not used in current build
2	CTSI_TRIG	Discrete	C21	Capacitive touch safety trip flag (SET by X7 rung)
3	CURRENT	Signed int 16	V3000	Scaled motor current (0–140 range, from WX0 via rung 3 SCALE)
4	D-MAN A	Discrete	C23	TESTMAN_A_OK — HMI soft deadman A (Test screen)
5	D-MAN B	Discrete	C24	TESTMAN_B_OK — HMI soft deadman B (Test screen)
6	DEADMEN_OK	Discrete	C0	Both deadman buttons (physical or test) held simultaneously
7	ERROR_LIGHT	Discrete	Y4	Error indicator output — physical panel fault LED

8	ERROR_OFF	Discrete	C18	Error reset coil — clears all latched faults when momentarily SET
9	ERROR_SCREEN	BCD int 16	V100	HMI screen selector — written to 3 on fault, 0 after 3 s (T4)
10	E_STOP_TRIG	Discrete	C26	E-stop triggered flag — SET when X4 (STOP_PB) opens
11	FREQ	Signed int 16	V11	VFD output frequency in 0.1 Hz units (read from VFD via serial or calc)
12	FWD	Discrete	C8	FORWARDS_SET — forward direction selected (latching bit)
13	GRAPHIC IND 1	Discrete	—	HMI-local graphic indicator; unused
14	JAMMED_MOTOR	Discrete	C20	Motor jammed and locked out (SET by CT0.Done rung)
15	JAM_COUNTER	BCD int 16	N0	CT0 accumulator — jam count 0–3 (N0 = CT0.ACC in Do-more)
16	MOTOR_STATUS	Discrete	C12	RUN_REQUEST — motor run command active
17	OVERCURRENT	Discrete	C25	OVERCURRENT_TEST — soft overcurrent trigger (Test screen button)
18	RPM	Signed int 16	V14	Calculated or measured shaft RPM (destination register)
19	RPM_CTRL	Signed int 16	V2000	Speed setpoint (0–1600 counts) — scaled to WY0 (0–4095) for VFD analog out

20	RVS	Discrete	C9	REVERSE_SET — reverse direction selected (latching bit)
21	SPEED	Signed int 16	—	HMI-local; unused
22	SPEED(1)	Signed int 16	—	HMI-local; unused
23	STOP	Discrete	—	HMI-local; unused
24	STOP_REQ	Discrete	C1	HMI_STP_REQ — operator stop request from HMI STOP button
25	TEMP	Signed int 16	V15	Temperature register (destination, RTD module — not yet calibrated)
26	VALUE	Signed int 32	—	HMI-local; unused
27	VOLTAGE	Signed int 16	V13	Voltage register (destination — source TBD)

6.2.2 Screen 1 — MOTOR CONTROL

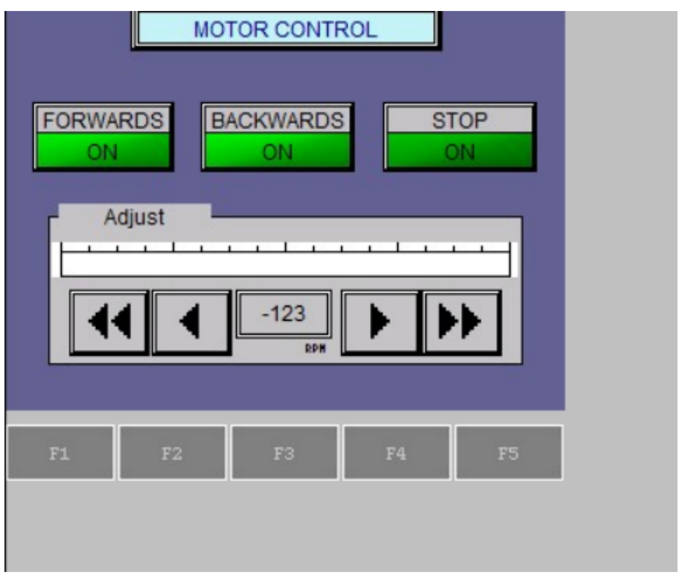

Figure 9: HMI Screen 1 — Motor Control (operator screen).

The primary operator screen. Header bar reads **MOTOR CONTROL**. Contains three toggle pushbuttons and a speed adjust panel.

- **FORWARDS button** (`FWD / C8`, toggle): toggles FORWARDS_SET. ON = green, OFF = dark. Sets C8, resets C9 (rung 13).
- **BACKWARDS button** (`RVS / C9`, toggle): toggles REVERSE_SET. Sets C9, resets C8 (rung 14).
- **STOP button** (`STOP_REQ / C1`, toggle): toggles HMI_STP_REQ. Rung 7 uses NOT C1 to block RUN_REQUEST.
- **Bar graph** (`RPM_CTRL / V2000`): horizontal bar 0–1600 showing current V2000 value.
- **Numeric display** (`RPM_CTRL / V2000`): 3-digit integer labeled 'RPM' below.
- **IncDec ±50 / ±250** (`RPM_CTRL / V2000`): fine and coarse speed adjust buttons.
- **F1/F2/F3 keys**: navigation. F1 → Screen 1, F2 → Screen 2, F3 → Screen 3.

Speed control data path: V2000 (RPM_CTRL, 0–1600) is scaled by rung 2 to WY0 (0–4095), which drives the analog output module F2-08DA-2 CH1 → VFD VI terminal (0–10 V frequency reference).

6.2.3 Screen 2 — MOTOR TELEMETRY

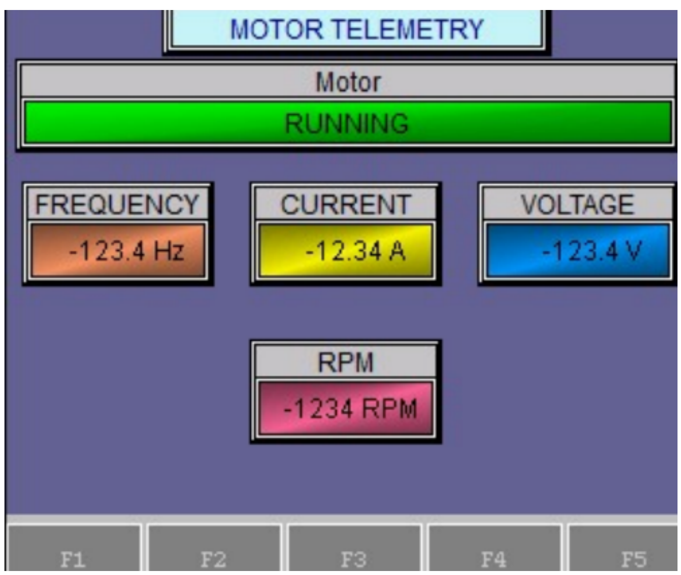

Figure 10: HMI Screen 2 — Motor Telemetry (read-only status).

Header bar reads **MOTOR TELEMETRY**. Read-only status display. A full-width indicator light spans the top and shows RUNNING (green) or STOPPED (gray).

- **Motor status indicator** (`MOTOR_STATUS / C12`): full-width bar, ON = 'RUNNING' green, OFF = 'STOPPED' gray.
- **FREQUENCY display** (`FREQ / V11`): 4-digit, 1 decimal, Hz. Orange background.
- **CURRENT display** (`CURRENT / V3000`): 4-digit, 2 decimal, A. Yellow background.
- **VOLTAGE display** (`VOLTAGE / V13`): 4-digit, 1 decimal, V. Blue background.
- **RPM display** (`RPM / V14`): 4-digit, 0 decimal, RPM. Pink background.
- **F1/F2/F3 keys**: navigation. F3/F4/F10 → Screen 3.

6.2.4 Screen 3 — ERROR (Fault / Alarm Screen)

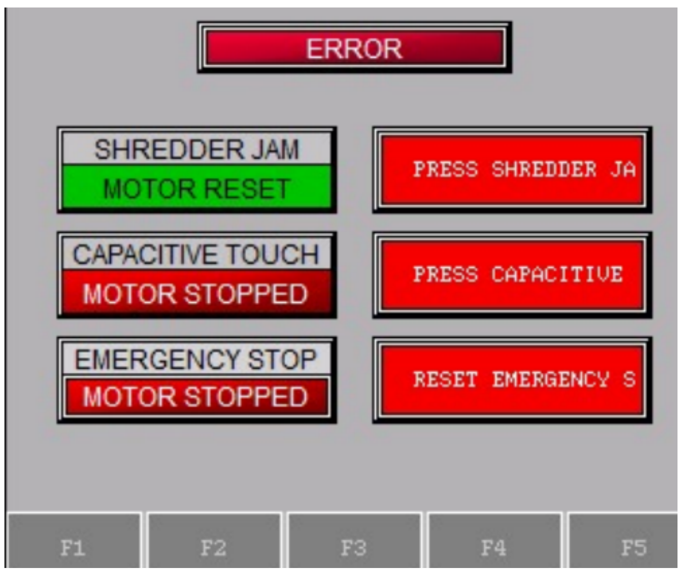

Figure 11: HMI Screen 3 — Error / Alarm screen (red backlight, three fault rows).

Backlight color 11 (red backlight). Automatically shown when CHANGE_SCREEN (C22) is SET by the ladder (rung 29 writes V100=3; after 3 s T4 resets V100 to 0). Three fault rows each with a status indicator and scrolling instruction text.

- **ERROR header indicator** (`ERROR_LIGHT / Y4`): ON shows 'ERROR' in red; OFF shows 'ERROR RESET' in gray. Y4 is the physical fault LED output.
- **SHREDDER JAM row** (button + indicator): button `ERROR_OFF / C18`; indicator `JAMMED_MOTOR / C20`. Indicator lit red ('MOTOR STOPPED') when C20 HIGH (CT0.Done fired). Button pulses C18 (ERROR_RESET).

- **CAPACITIVE TOUCH row** (button + indicator): button `ERROR_OFF / C18`; indicator `CTSI_TRIG / C21`. Indicator lit red when C21 HIGH (capacitive touch trip).
- **EMERGENCY STOP row** (indicator only): `E_STOP_TRIG / C26`. ON shows 'MOTOR STOPPED' red; OFF shows 'MOTOR RESET' green.
- **Scroll text**: each row scrolls instructions: 'PRESS SHREDDER JAM BUTTON TO RESET MOTOR.', 'PRESS CAPACITIVE TOUCH TRIGGER BUTTON TO RESET MOTOR.', 'RESET EMERGENCY STOP BUTTON TO RESET MOTOR.'

6.2.5 Screen 4 — TEST (Engineering / Commissioning)

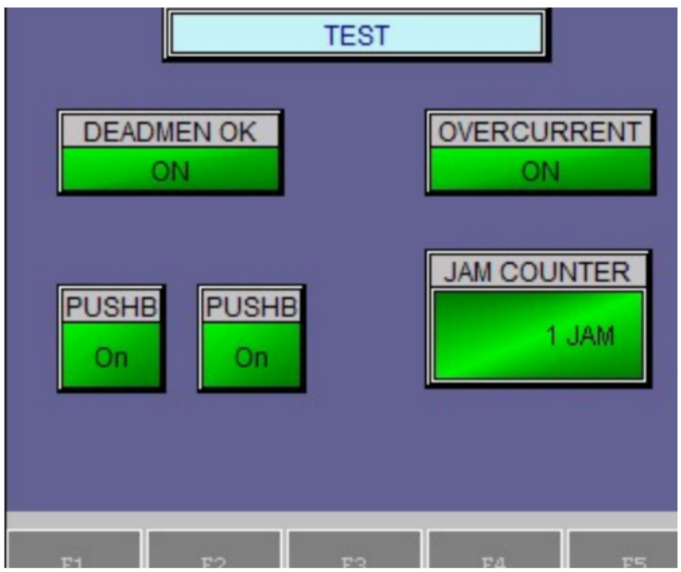

Figure 12: HMI Screen 4 — Test / Commissioning (soft inputs for bench testing).

Header reads **TEST**. Used during commissioning to simulate inputs without the physical pushbuttons. Not accessible from normal operation screens (only reachable via F5 from Screen 3).

- **DEADMEN OK button** (`DEADMEN_OK / C0`, toggle): directly forces C0 HIGH. Allows testing run logic without holding physical buttons.
- **OVERCURRENT button** (`OVERCURRENT / C25`, momentary on): pulses C25 (OVERCURRENT_TEST). Rung 19 treats C25 same as X5, triggering the full jam recovery sequence.
- **JAM COUNTER display** (`JAM_COUNTER / N0`, BCD): shows CT0 accumulator as jam count 0–3.

- **PUSHBUTTON A** (`D-MAN A / C23`, toggle): forces C23 (TESTMAN_A_OK) HIGH. Rung 5 uses C23 as alternate to X2 for deadman A.
- **PUSHBUTTON B** (`D-MAN B / C24`, toggle): forces C24 (TESTMAN_B_OK) HIGH. Rung 5 uses C24 as alternate to X3 for deadman B.

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.
- Control wires land as straight stripped conductors at the terminal blocks (no ferrules used in this build).

7.2 Safety Functions

- **E-stop:** Category 0 stop (immediate removal of power). Hardwired — an E22B1 NC contact in series with the SD-N35 contactor coil; pressing it drops the coil and opens the contactor independently of the PLC. Single channel; the PLC additionally monitors the state on X4 (not a safety-rated channel).
- **Two-hand / deadman control:** both blue pushbuttons must be held continuously to enable the motor. As-built, the two NO inputs are AND-ed in the standard PLC's ladder logic to gate the VFD run output. It is the run-permissive and the safety interlock satisfying requirement M6, but it is *not* a safety-rated two-hand device per ISO 13851 Type III (no simultaneity timer, complementary contacts, or self-monitoring).
- **Access guarding:** handled by the ME team's mechanical design (feed geometry), so no electrical lid interlock is implemented (a lid interlock was a may-have, not a requirement).
- **Capacitive-touch module (planned):** a capacitive-touch safety module is on hand and the ladder logic to act on it (input X7 → stop motor + fault indication) is written, but it is *not yet wired*. This is a may-have (additional safety sensing) and is not relied upon in the current build.

- **Overload:** the VFD's electronic motor-overload function plus the over-torque trip (see Jam Detection); equipment protection rather than a personnel-safety function.

7.3 Performance Level (ISO 13849-1) — Preliminary Assessment

This is a preliminary, self-assessed determination from the as-built architecture. A formal verification needs a documented ISO 12100 risk assessment (to set the required performance level, PLr) and component reliability data (MTTFd / B10d, diagnostic coverage). Table 2 summarizes the achieved categories.

Table 2: Preliminary ISO 13849-1 assessment of the as-built safety functions.

Safety function	**As-built architecture**	**Category (prelim.)**
Emergency stop	Hardwired E22B1 NC contact in series with the SD-N35 contactor coil; opening it removes motor power. Single channel, well-tried components; PLC monitors X4 (not safety-rated).	Cat 1 (PL b–c)
Two-hand / deadman	Two NO buttons AND-ed in the standard PLC to gate the VFD run output. Single channel; no simultaneity check, complementary contacts, or self-monitoring.	Cat B (PL b); not ISO 13851 Type III
Over-torque / jam stop	VFD over-torque trip plus electronic motor overload. Equipment protection, not a personnel-safety function.	Not rated (process protection)

Overall the control system is approximately **Category B–1 (PL b–c)**, limited by: single-channel architecture throughout; a standard (non-safety) PLC evaluating the two-hand permissive; a single contactor with no force-guided (mechanically linked) contacts and no feedback (EDM) monitoring to detect a welded contactor; and the as-built neutral-switching deviation (see the NEC note). The hardwired E-stop, built from well-tried components, is the strongest function (Category 1).

To raise the achieved PL — for example to Category 3 / PL d, should the risk assessment require it for the rotating-blade hazard — would need a dual-channel architecture, a safety relay or safety-rated PLC, two contactors (or force-guided contacts) with feedback monitoring, switching of the hot leg, and a synchronous two-hand module meeting ISO 13851.

[TODO: Confirm with the advisor whether formal PL/SIL verification is in capstone scope. If so, document the target PLr from the ISO 12100 risk assessment and the component MTTFd / B10d and DC values to complete the calculation.]

7.4 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.
- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 User Manual

8.1 Purpose of This Manual

This manual provides operators, maintenance personnel, and safety officers with the procedures necessary to safely operate, monitor, and perform basic troubleshooting on the Universal Plastic Shredder V2.0 electrical control system. It assumes the user has read the full engineering report and understands the basic hazards described in Section 7 (Safety Design).

8.2 Intended Audience and Required Training

- **Operators:** Must complete a site-specific safety briefing covering two-hand deadman operation, E-stop use, and jam recovery. No electrical troubleshooting is permitted without additional training.
- **Maintenance personnel:** Must be familiar with NFPA 70E arc-flash boundaries, lockout/tagout (LOTO) per 29 CFR 1910.147, and the contents of Appendix B (I/O point list) and Appendix D (VFD parameters).
- **Supervisors / safety officers:** Shall maintain a copy of this manual and the full engineering report adjacent to the machine or in a controlled electronic repository.

8.3 Normal Operating Procedure

8.3.1 Pre-Start Checks

1. Verify the main disconnect (PowerTec 71008) is in the OFF position.
2. Inspect the feed area for obstructions, loose plastic, or foreign objects.
3. Verify the E-stop button is pulled out (released).
4. Ensure both deadman pushbuttons are not pressed.
5. Close and latch any mechanical access guards (ME team's responsibility).

8.3.2 Startup

1. Turn the main disconnect to ON. The HMI shall display **IDLE / READY**. The green status indicator shall illuminate.
2. If the HMI shows a fault or lockout message, press the HMI **RESET** button (see Section 8.6).

3. To start shredding:
   - Press and **hold both blue deadman pushbuttons simultaneously**.
   - The motor will accelerate to the programmed speed (70–90 RPM at the shaft).
   - The red "RUNNING" status indicator shall illuminate.
   - If either deadman button is released, the motor stops immediately.
4. Feed plastic sheet material at a moderate, steady rate. Avoid overstuffing.

8.3.3 Normal Shutdown

1. Release both deadman buttons. The motor will coast to a stop.
2. Wait for the blades to come to a complete stop (minimum 5 seconds).
3. Turn the main disconnect to OFF.

8.4 Emergency Stop and Safety Functions

- **To stop in an emergency:** Press the red mushroom E-stop button. Motor power is removed immediately (Category 0 stop). The HMI will show **ESTOP ACTIVE**.
- **To reset after E-stop:** Twist the E-stop button clockwise to release it, then press the HMI RESET button.
- **Two-hand deadman:** The motor runs only while both blue buttons are held. This is a permissive — not a safety-rated two-hand control per ISO 13851 Type III. Never bypass or defeat this function.

8.5 Jam Detection and Automatic Recovery

The control system includes an auto-reverse jam recovery feature. When an over-torque jam is detected (motor current exceeds 150% of rated for 3 seconds):

1. The VFD trips and the motor stops.
2. The HMI displays **JAM DETECTED** and increments the jam counter.
3. The PLC automatically:
   (a) Sends a 500 ms reset pulse to clear the VFD fault.
   (b) Runs the motor in reverse for 2 seconds to clear the jam.

(c) Pauses briefly, then restarts forward operation.

4. If the jam clears, normal operation resumes.
5. If a jam occurs again within the same operating session:
   - Jam counter increments.
   - After the **third jam** within a session, the system enters **LOCKOUT** mode.
   - The HMI shows **LOCKOUT — MANUAL RESET REQUIRED**.
   - Motor cannot restart until a supervisor presses the HMI RESET button and clears the feed area.

8.6 Fault Recovery and Lockout Reset

Table 3: Fault codes and recovery actions

HMI Display	Operator Action
ESTOP ACTIVE	Release E-stop button, then press HMI RESET.
VFD FAULT	Check VFD display for code (overvoltage, overtemp, etc.). Power cycle main disconnect. If repeated, call maintenance.
JAM (1 or 2)	No action — automatic recovery runs. Monitor HMI.
LOCKOUT (3 jams)	Clear plastic from blades manually (after LOTO), then press HMI RESET.
IDLE / READY	Normal.

8.7 Schematic and Wiring Diagrams

The following figures show the core electrical circuits of the shredder control system. All wire colors follow NFPA 79 §13.2 (see Section 7.1). Full terminal-level details are in Appendix B (I/O point list).

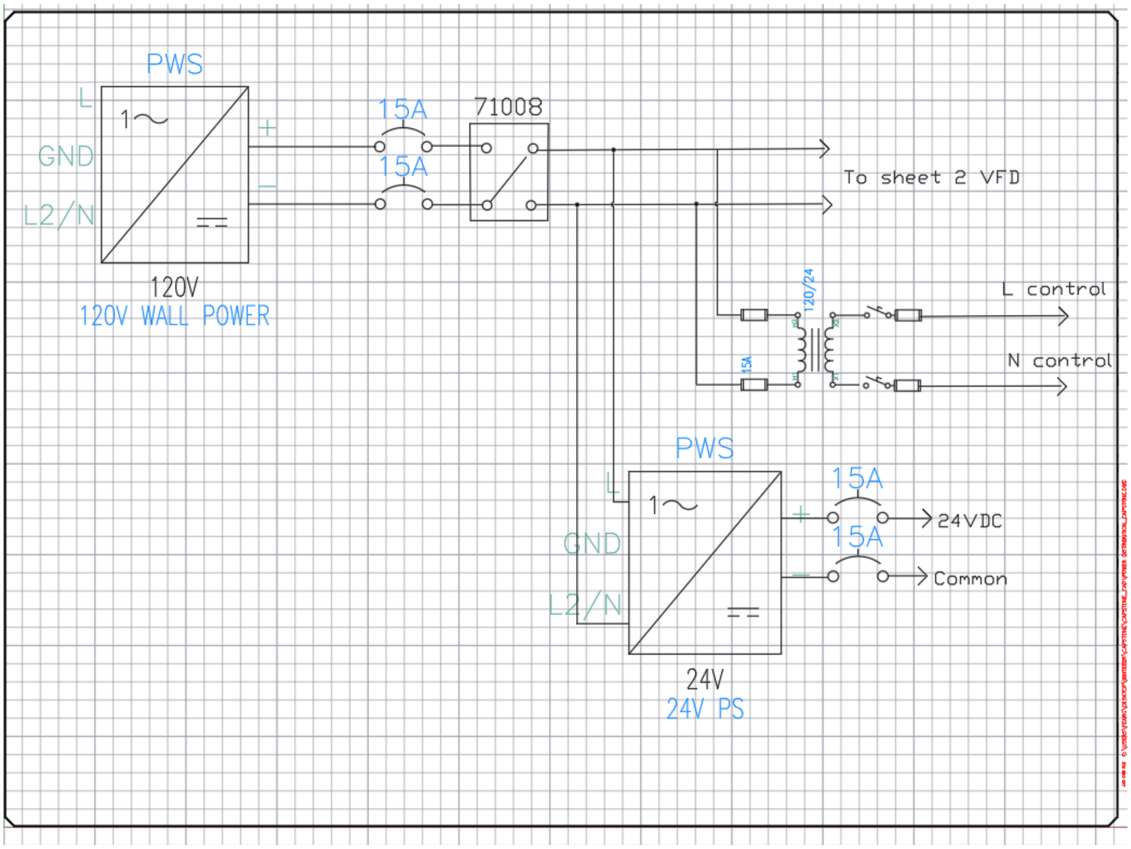

Figure 13: Power distribution circuit: main disconnect, branch breakers, 24 V power supply, and VFD input.

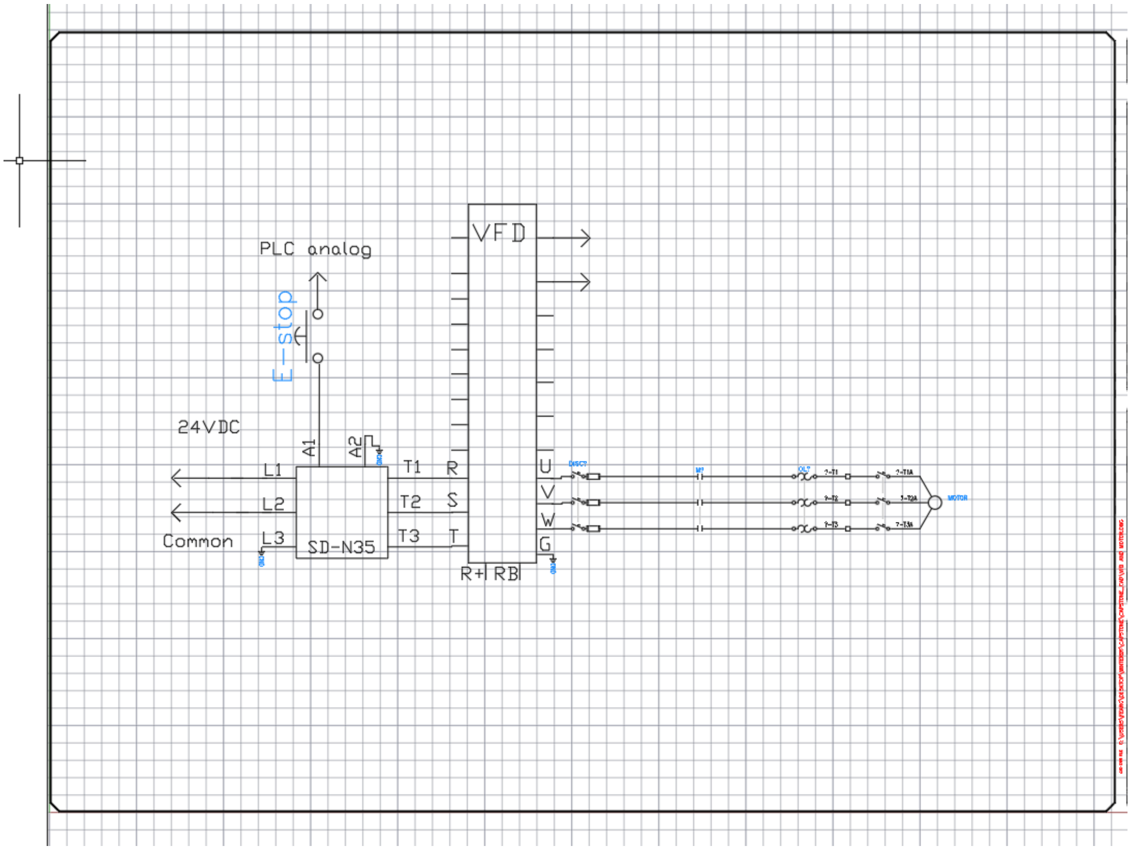

Figure 14: Motor control circuit: VFD connections to motor, brake resistor (P/Pr), and fault relay output (FA/FC).

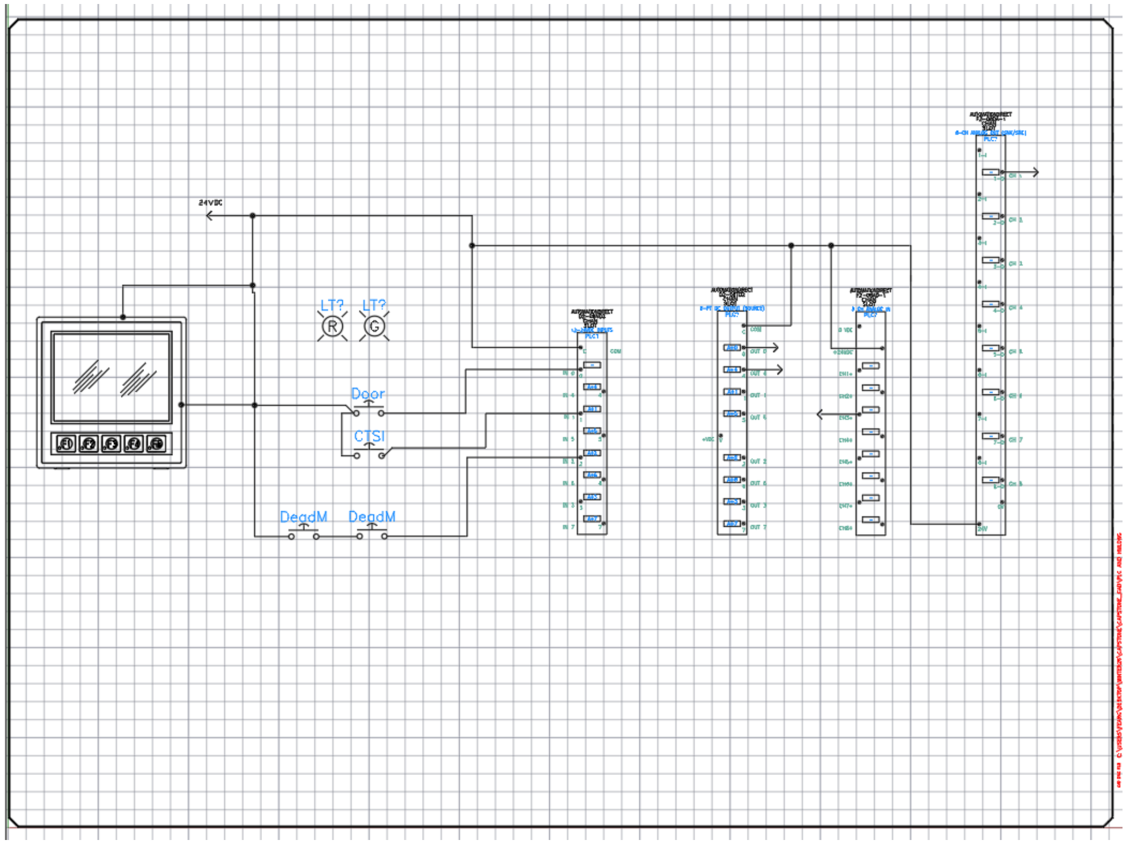

Figure 15: PLC I/O circuit: deadman buttons, E-stop monitor, VFD fault input, and VFD run/rev/reset outputs.

8.8 Daily and Periodic Maintenance

8.8.1 Operator Daily Checks

- Before first use: Verify E-stop button stops the motor immediately.
- Check that both deadman buttons are responsive (motor runs only when both held).
- Listen for unusual noises (grinding, excessive vibration).
- Ensure the HMI jam counter resets to zero after a clean 30-second run.

8.8.2 Monthly Maintenance (Qualified Personnel Only)

- Perform LOTO (lockout/tagout) on the main disconnect.
- Open the electrical enclosure.
- Visually inspect for loose wires, discoloration, or burned terminals.

- Check that the brake resistor (75 ohm, 500w) shows no cracking or overheating signs.
- Verify fan operation on the VFD and power supply.
- Clean dust from the enclosure using low-pressure compressed air.
- Close and secure enclosure before re-energizing.

8.9 Troubleshooting Guide

Table 4: Common issues and corrective actions

Symptom	Possible cause	Action
Motor does not start	E-stop pressed	Release E-stop, reset HMI.
	Only one deadman pressed	Press both simultaneously.
	LOCKOUT active	Clear jam, press HMI RESET.
Motor runs briefly then stops	Jam detected	Let auto-reverse run. If repeated, clear plastic manually.
HMI blank	No 24V power	Check Mean Well PSU input breaker (10A) and output fuses.
VFD shows fault code	Overvoltage / overtemp / etc.	See VFD manual Appendix D. Power cycle.
Jam counter increments without jam	Over-torque threshold too low	Check PD124. Nominal = 150%.

8.10 Safety Warnings Summary

CRITICAL SAFETY WARNINGS

- Never reach into the feed area while blades are moving. Wait for complete stop.
- Do not bypass, tape down, or modify the deadman buttons or E-stop.
- Lockout/tagout the main disconnect before any enclosure opening or blade service.
- This machine has no electrical feed interlock — the mechanical feed geometry is the primary access safeguard.
- If the automatic jam recovery cycles three times, **stop and manually clear** the jam.

8.11 Contact for Service and Support

For maintenance beyond this manual, contact the Ichor Systems facilities team or the original design team (via Portland State University, ECE Department). Have the following ready:

- Shredder serial number (TBD — assign upon final assembly)
- VFD fault code (if any)
- HMI jam count and lockout status
- Photos of any visible damage

8.12 Revision History of This Manual

Version	Date	Changes
1.0	2026-05-31	Initial release (rough draft).

9 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

9.1 Standards Summary Matrix

Standard	Scope	How it is applied here
NFPA 70 (NEC)	National Electrical Code – premises wiring, motor circuits	Branch-circuit, conductor, grounding, and disconnect design (see §9.2).
NFPA 79	Electrical standard for industrial machinery	Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §9.3).
NEMA 250	Enclosure type ratings	Control enclosure selected to a NEMA type matched to the lab/shop environment (see §9.4).
NEMA ICS 1/2	Industrial control & systems	Contactor / control device selection and marking.
NEMA MG 1	Motors and generators	Motor nameplate interpretation and VFD parameterization.
UL 508A	Industrial control panels	Panel construction practice and short-circuit current rating (SCCR) awareness.
IEC 60204-1	Safety of machinery – electrical equipment	International counterpart to NFPA 79; used to cross-check stop categories.
ISO 12100	Risk assessment / risk reduction	Framework behind the hazard analysis driving the safety circuit.
ISO 13849-1	Safety-related parts of control systems (PL)	Preliminary self-assessment (Cat B–1) of the E-stop and two-hand control chain.
ISO 13850	Emergency stop function	E-stop actuator type, color, and Category 0 behavior.
ISO 13851	Two-hand control devices	Deadman two-hand start timing and configuration.
OSHA 29 CFR 1910.147	Lockout/Tagout (LOTO)	Lockable main disconnect for energy isolation during service.

9.2 NFPA 70 – National Electrical Code

NEC article	Requirement	How it is applied here
Art. 430.6	Motor FLC basis	From the GE test-motor nameplate (FLA 7.6 A at 230 V); basis for all downstream sizing. Re-verified when the production motor is selected.
Art. 430.22	Branch conductor sizing	Conductors sized at $\geq$ 125% of motor FLC.
Art. 430.32	Running overload	VFD electronic motor-overload (`PD142 = 7.6 A`) plus the over-torque trip.
Art. 430.52	Branch short-circuit / ground-fault	30 A C-curve breaker on the VFD branch; clears inrush without nuisance trips (inverse-time-breaker allowance).
Art. 430.102/.109	Disconnecting means	Provided ahead of the controller and VFD.
Art. 430 Part X (430.122)	Adjustable-speed drive systems	Conductor and protection rules applied to the VFD-fed motor circuit.
Art. 250	Grounding & bonding	EGC (green / green-yellow) bonds the VFD, PSU, motor frame, and panel to a common ground bus and building earth.
Art. 310	Conductor ampacity	Conductors chosen from the ampacity tables for the installed temperature rating.
Art. 409	Industrial control panels	Panel layout, marking, and overcurrent considerations.
Art. 110.26	Working space	Clearances around live parts maintained for safe service. **[TODO: Confirm final panel mounting leaves required clearance.]**

[NOTE: As-built deviation flagged during the Point List review: the SD-N35 safety contactor switches the VFD neutral *(T) leg rather than the* hot *(R) leg. NEC / NFPA 79 practice is to switch the ungrounded (hot) conductor so that opening the contactor de-energizes the load to ground potential. Switching the grounded conductor leaves the load at line potential when the contactor is open. This is documented in the as-built I/O point list (Appendix B,*

VFD terminals) and must be corrected, or explicitly justified and risk-assessed, before final sign-off.]

9.3 NFPA 79 – Electrical Standard for Industrial Machinery

NFPA 79 clause	Requirement	How it is applied here
§5.3	Supply disconnecting means	Lockable main disconnect (PowerTec 71008) isolates the machine from supply; also satisfies LOTO.
§7	Overcurrent protection	Branch and downstream fusing at terminal blocks (PSU 25 A slow-blow, PLC 1 A fast-blow, control loads individually fused).
§8	Protective bonding circuit	Continuous equipment-grounding / bonding path to all exposed conductive parts.
§9	Control circuits & functions	Stop functions classified by stop category; the hardwired safety chain has precedence over the PLC.
§10.7–10.8	Operator interface / E-stop	Red mushroom actuator on a yellow background, self-latching NC contact; start devices guarded by the two-hand requirement.
§13.2	Conductor color coding	Black AC line/load, red AC control, blue DC control, blue/white DC return, green/yellow PE; wires labeled both ends, landed straight (no ferrules in this build).

9.4 NEMA Standards

Standard	Covers	How it is applied here
NEMA 250	Enclosure type ratings	Targeting Type 12 (dust-tight, drip-tight) for the PLC/HMI panel. **[TODO: Confirm final enclosure type and dust-ingress handling.]**
NEMA ICS 1 / ICS 2	Industrial control & systems	Contactor and control-device selection/marking; the Mitsubishi SD-N35 is sized above motor FLA for AC-3 (motor) duty.
NEMA MG 1	Motors & generators	Interpreting the motor nameplate (voltage, poles, RPM, service factor) for VFD parameters `PD141--PD144`.

9.5 Machine-Safety Standards (ISO / IEC)

Standard	Covers	How it is applied here
ISO 12100	Risk assessment / reduction	Identifies the primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.
ISO 13849-1	Safety-related control parts (PL)	Preliminary assessment: hardwired E-stop $\approx$ Cat 1; two-hand permissive $\approx$ Cat B (see the Safety Functions section).
ISO 13850	Emergency stop function	Category 0 stop (immediate removal of power) with fail-safe NC wiring.
ISO 13851	Two-hand control devices	Both buttons actuated within 0.5 s of each other and held continuously to enable the motor.
IEC 60204-1	Electrical equipment of machines	International analogue of NFPA 79; used to cross-check stop categories and supply disconnection.

9.6 Occupational Safety (OSHA)

Regulation	Covers	How it is applied here
29 CFR 1910.147	Lockout/Tagout (LOTO)	The main supply disconnect is lockable for positive isolation during blade-area service and maintenance.
29 CFR 1910 Subpart S	Electrical safety	General electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build.

[TODO: Before final submission: convert any remaining "targets / intends to" language above into verified compliance statements, attach the panel enclosure datasheet, and resolve the NFPA 79 neutral-switching deviation note.]

10 Testing and Validation

10.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (`PD124 = 1`, `PD125 = 5.0`): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (`PD124 = 150`) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

10.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

10.3 Safety Validation

- E-stop response time: **[TODO: measure with scope]**.
- Two-hand control timing: **[TODO: verify both buttons must be held continuously; simultaneous release stops within X ms]**.

11 Results

11.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with the PLC over the serial COM link.

11.2 Open Issues

- Vendor-specific Huanyang firmware quirk: `PD052 = 12` does not fire relay in continue-running mode. Working around with `PD052 = 02` + `PD123 = 3`.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.
- Motor-current monitor (VFD `VO` → F2-08AD-1 CH1) is wired, but under time constraints it was never configured/scaled correctly to read the VFD output current. The signal path exists; the VFD analog-output and PLC analog-input scaling still need to be set and calibrated.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

12 Discussion

12.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD ($150) over ATO ($450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.

- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NO active-high versus NC fail-safe jam signal: the VFD fault relay is wired NO active-high (FA–FC closes on trip, X5 HIGH). Simpler to wire, but not fail-safe at the signal level—a broken wire reads as "no fault." That residual risk is covered by the independent hardwired E-stop loop and motor contactor, which remove power regardless of the PLC.

12.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on control wires in future builds: this build landed straight stranded wire at the terminal blocks, and loose/spread strands caused multiple debugging hours during commissioning.

12.3 Project Management Notes

[TODO: Add a paragraph on how the team coordinated, what tools we used (GitHub, weekly meetings, etc.), and how schedule slipped or held.]

13 Future Work and Improvements

13.1 Safety and Compliance

- **Integrate the capacitive-touch safety (CTSI) module:** wire the on-hand capacitive-touch module to input X7. The ladder logic to stop the motor and raise a fault on a touch trip is already written, so this is primarily a wiring and validation task. It adds a no-contact operator-presence safeguard at the feed.
- **Raise the safety performance level:** move from the current single-channel Cat B–1 toward Cat 3 / PL d — a safety relay or safety-rated PLC, dual-channel inputs,

two contactors (or force-guided contacts) with feedback (EDM) monitoring, and a synchronous two-hand module per ISO 13851.

- **Correct the supply switching:** switch the hot (ungrounded) conductor through the contactor rather than the neutral, per NFPA 79 / NEC.
- **Ferrule the control wiring:** crimp ferrules on stranded control conductors at the terminal blocks to eliminate the loose-strand faults seen during commissioning.

13.2 Monitoring and Connectivity

- **Telemetry:** stream live operating data (motor current, speed, jam count, fault state, runtime) off the PLC for dashboards and trend analysis.
- **Modbus to the VFD:** the Huanyang supports RS-485 Modbus (currently unused). Adding it would pull far richer data straight from the drive — output current, voltage, frequency/RPM, DC-bus voltage, drive temperature, and detailed fault codes — beyond the single 0–10 V analog monitor.
- **Internet / remote viewing:** expose a read-only view of device status over the network — via the H2-DM1E's built-in web server or an ESP32 / edge gateway — so the shredder can be monitored remotely from a browser or phone.
- **Data logging:** record total runtime, jam events per shift, and faults locally for predictive maintenance.

13.3 Capability

- **Predictive jam detection:** train a small model on the motor-current waveform shape just before a jam to detect and pre-empt jams earlier.
- **Mechanical integration:** add a feed hopper that auto-meters plastic into the blades.
- **Higher-capacity model:** scale to a 5–10 HP drivetrain for greater throughput.

14 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – layered safety functions (hardwired E-stop, two-hand deadman control), NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.

- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

15 References

- NFPA 70 (2023): National Electrical Code (esp. Articles 110, 250, 310, 409, 430).
- NFPA 79 (2024): Electrical Standard for Industrial Machinery.
- NEMA 250: Enclosures for Electrical Equipment.
- NEMA ICS 1 / ICS 2: Industrial Control and Systems.
- NEMA MG 1: Motors and Generators.
- UL 508A: Industrial Control Panels.
- IEC 60204-1:2016: Safety of Machinery – Electrical Equipment of Machines.
- ISO 12100:2010: Safety of Machinery – Risk Assessment and Risk Reduction.
- ISO 13849-1:2023: Safety-related parts of control systems.
- ISO 13850:2015: Emergency Stop Function.
- ISO 13851:2019: Two-hand control devices.
- OSHA 29 CFR 1910.147: The Control of Hazardous Energy (Lockout/Tagout).
- Huanyang HY-series VFD User Manual.
- AutomationDirect DL205 / Do-more User Manual.
- GE 5KE Frame 182T Motor Datasheet.

[TODO: Add full reference list with URLs / publication info before final.]

16 Appendices

16.1 Appendix A: Bill of Materials

The complete as-built Bill of Materials is consolidated in `BOM/BOM_Combined_As_Built.xlsx` (and the rendered `BOM/BOM_Combined_As_Built.pdf`) in the project repository. The workbook contains three sheets:

- **Combined BOM** — every item actually installed in the control panel, grouped by category (Motor & Drive, Power Distribution, PLC Control System, HMI, Operator Controls, Motor Connections, Wire/Duct/Hardware) with manufacturer, part number, quantity, and notes pointing back to the relevant documentation.
- **Excluded - Not Used** — items that appeared in preliminary BOMs but were not used in the final build, with the reason (rejected during selection, not purchased, superseded by another component, or out of project budget).
- **VFD Decision Matrix** — the weighted comparison used to select the Huanyang HY02D211B-T (price 50%, current capability 20%, build quality 15%, documentation 10%, brand 5%).

Critical items (highlighted in the workbook): Motor (GE 5KE182BC205B for bench test), VFD (Huanyang HY02D211B-T), Brake Resistor (ATO APCS-300R30-AD), Mean Well NDR-480-24 24V DC power supply, Mitsubishi SD-N35 contactor, Cutler-Hammer E22B1 E-stop, AutomationDirect PLC rack with H2-DM1E CPU and the D2-08ND3 / F2-08AD-1 / F2-04RTD / D2-08TR / F2-08DA-2 modules, EA1-T4CL HMI.

[NOTE: The bench-test motor is the GE 5KE182BC205B (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM) used to validate the control system. The final production motor remains TBD per the 3-5 HP project requirement; its nameplate will supersede VFD parameters `PD141`–`PD144`. Control-panel detail (rack layout, terminal-by-terminal wiring) is itemized in the point list in Appendix B.]

16.2 Appendix B: PLC I/O Point List

The complete as-built point list, transcribed from the project workbook (`Point List.xlsx`), follows on the landscape pages below — every sheet, using the spreadsheet's columns *Point Description, Origin Address, DO / DI / AO / AI / Pwr, Destination Address, Destination Description, and Notes*: the index; the Mitsubishi SD-N35 contactor; the Mean Well NDR-480-24 power supply; the DirectLOGIC 205 rack and the H2-DM1E CPU; the EA1-T4CL HMI; every I/O module (D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2), the legacy D2-08TA and F2-02DAS-2 sheets; and the Huanyang HY02D211B-T VFD.

Sheet: Index

POINT ASSIGNMENT INDEX					
	PAGE NO.	NO. OF SHEETS	DESCRIPTION		
	1	1	Mitsubishi SD-N35		
	2	2	Mean Well NDR-480-24		
	3	3	Direct Logic 205 D2-09B-1		
	REV	DATE	DESCRIPTION	Created By	Checked by
	1				
	2				
	2				
	2				
	3				
	4				

Sheet: Mitsubishi SD-N35

Point Description	**Origin**	**DO**	**DI**	**AO**	**AI**	**Pwr**	**Destination Address**	**Destination Description**	**Notes**
Contactor (Mitsubishi SD-N35) – Coil A1	A1	0	0	0	0	1	Y0	D2-08TD2 Output Module (12–24VDC)	A1 $\rightarrow$ DF103V (1/2A slow blow fuse) $\rightarrow$ D2-08TD2 output supplying +24VDC
Contactor (Mitsubishi SD-N35) – Coil A2	A2	0	0	0	0	1	-V	NDR-480-24 Power Supply	A2 $\rightarrow$ direct return to power supply negative (–V)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Contactor (Mitsubishi SD-N35) – Line Input	L1	0	0	0	0	1	L	Phoenix Contact TMC 83C 15A (2907618)	L1 $\rightarrow$ Phoenix Contact fuse breaker $\rightarrow$ fed from wall AC line
Contactor (Mitsubishi SD-N35) – Load Output	T1	0	0	0	0	1	R	HY Series VFD (HY02D211B-T)	T1 $\rightarrow$ passes through contactor $\rightarrow$ supplies VFD terminal R (line input)
	Total Points								

Sheet: Mean Well NDR-480-24

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
		0					AC Mains Hot from wall via main power switch and 25 A fuse	120 VAC line conductor from wall outlet, routed through panel main disconnect, then through 25 A slow-blow fuse on EURO S10H-5H fuse block	Wall AC L -> main disconnect -> EURO S10H-5H 25 A slow-blow -> PSU L. Black wire per NFPA 79.
Mean Well NDR-480-24 N (AC Neutral Input)	N	0	0	0	0	1		120 VAC neutral conductor from wall outlet, returns through neutral bus	Wall AC N -> neutral bus -> PSU N. White wire per NFPA 79.
Mean Well NDR-480-24 G (Ground / PE)	G	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds PSU chassis to panel ground bus and ultimately to building earth	PSU G stud -> panel ground bus. Green or Green/Yellow wire per NFPA 79.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Common)	+V	0	0	0	0	1	EURO S4LH Terminal Block -> Pushbutton Commons	Fused +24V distribution to operator pushbutton common via S4LH	+V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3B x2, Reset S2PR-P3R).
Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit	+V	0	0	0	0	1	Cutler-Hammer E22B1 NC contact + SD-N35 contactor coil A1	Direct +24V (no fuse) into E-stop safety chain	+V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this line between E22B1 and A1 to monitor E-stop state.
Mean Well NDR-480-24 +V to VFD FA (Jam Signal)	+V	0	0	0	0	1	VFD Huanyang HY02D211B-T FA terminal	Direct +24V (no fuse) into VFD jam signal circuit	+V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges and delivers +24V to PLC X5.
Mean Well NDR-480-24 +V to F2-08AD-1 (Module Power)	+V	0	0	0	0	1	F2-08AD-1 Analog Input Module +24V terminal	Direct module power	+V -> F2-08AD-1 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-08DA-2 (Module Power)	+V	0	0	0	0	1	F2-08DA-2 Analog Output Module +24V terminal	Direct module power	+V -> F2-08DA-2 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-04RTD (Module Power)	+V	0	0	0	0	1	F2-04RTD RTD Input Module +24V terminal	Direct module power	+V -> F2-04RTD +24V terminal. No fuse on this branch.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 +V to Indicator LEDs	+V	0	0	0	0	1	LED indicator (+) terminals (SA-LDG/Y/B/B/R, Wamco 525)	Direct +24V to (+) side of all illuminated push-button LED blocks and the separate Wamco error light	+V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when energized.
Mean Well NDR-480-24 -V to SD-N35 Contactor Coil A2	-V	0	0	0	0	1	Mitsubishi SD-N35 Contactor A2 (coil return)	Contactor coil return through E-stop chain	-V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop not pressed) and +V reaches A1.
Mean Well NDR-480-24 -V to D2-08ND3 (Input Common)	-V	0	0	0	0	1	D2-08ND3 Discrete Input Module Common (C)	Module input common for sourcing mode operation	-V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH when +24V is applied.
Mean Well NDR-480-24 -V to D2-08TR (Relay Common)	-V	0	0	0	0	1	D2-08TR Relay Output Module Common (C)	Relay common for sinking mode operation	-V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output terminal connects to -V (sinks current from the load).
Mean Well NDR-480-24 -V to F2-08AD-1 (Module Power Return)	-V	0	0	0	0	1	F2-08AD-1 Analog Input Module 0V terminal	Module power return and analog signal common	-V -> F2-08AD-1 0V terminal.
Mean Well NDR-480-24 -V to F2-08DA-2 (Module Power Return / Analog Common)	-V	0	0	0	0	1	F2-08DA-2 Analog Output Module 0V terminal	Module power return and analog signal reference for all 8 channels	-V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog signal common.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 -V to F2-04RTD (Module Power Return)	-V	0	0	0	0	1	F2-04RTD RTD Input Module 0V terminal	Module power return	-V -> F2-04RTD 0V terminal.
Mean Well NDR-480-24 -V to VFD DCM (Digital Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T DCM terminal	VFD digital input common reference	-V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/NPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM.
Mean Well NDR-480-24 -V to VFD ACM (Analog Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T ACM terminal	VFD analog signal common reference	-V -> VFD ACM. Shared 0V reference for the analog speed command (VI from F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).
Mean Well NDR-480-24 +V to HMI (EA1-T4CL)	+V	0	0	0	0	1	C-More EA1-T4CL HMI +VDC input	HMI panel power	+V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (replaces former Rhino PSB12-030-P).
Mean Well NDR-480-24 -V to HMI (EA1-T4CL)	-V	0	0	0	0	1	C-More EA1-T4CL HMI 0VDC return	HMI panel power return	-V -> HMI 0VDC. System DC common reference for the HMI panel.
Total Points									

Sheet: Direct Logic 205 D2-09B-1

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Base	Base	0	0	0	0	1	D2-09B-1	Base / Power Unit	Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Line Input	L	0	0	0	0	1	L	MORSETTITALIA EURO S10H-5H Fuse Block	Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal
PLC Rack (DirectLOGIC 205) – Neutral Input	N	0	0	0	0	1	N	Neutral Bus	Neutral from wall -> directly connected to PLC N terminal
PLC Rack (DirectLOGIC 205) – Ground (Protective Earth)	G	0	0	0	0	1	G	Ground Bus	Protective earth -> connected to PLC ground terminal / chassis bond
PLC Rack (DirectLOGIC 205) – Slot 1	Slot 1	0	0	0	0	0	H2-DM1E	CPU / Ethernet Module	CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.
PLC Rack (DirectLOGIC 205) – Slot 2	Slot 2	0	8	0	0	0	D2-08ND3	8-Point Discrete Input Module	8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors.
PLC Rack (DirectLOGIC 205) – Slot 3	Slot 3	0	0	0	8	0	F2-08AD-1	8-Channel Analog Current Input Module (4-20 mA)	8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).
PLC Rack (DirectLOGIC 205) – Slot 4	Slot 4	0	0	0	4	0	F2-04RTD	4-Channel RTD Input Module	4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Slot 5	Slot 5	8	0	0	0	0	D2-08TR	8-Point Relay Output Module	8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV).
PLC Rack (DirectLOGIC 205) – Slot 6	Slot 6	0	0	8	0	0	F2-08DA-2	8-Channel Analog Voltage Output Module (0-10 V)	8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).
PLC Rack (DirectLOGIC 205) – Slot 7	Slot 7	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 8	Slot 8	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 9	Slot 9	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.

Sheet: DirectLOGIC CPU (H2-DM1E)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Power and communication interface	Backplane Connector (Left Edge)	0	0	0	0	1	D2-09B-1 Base Slot 1 Connector	Provides power and backplane communication	–

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Ground reference	Backplane GND	0	0	0	0	1	D2-09B-1 Base Ground Bus	Common ground for logic power	–
HMI communication port	CPU Comm Port	0	0	0	0	0	HMI EA1-T4CL COM Port	4" C-More Micro-Graphic panel (operator interface)	CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel.
	Total Points								

Sheet: Output Module (D2-08TA)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Output Common (Hot)	C	1	0	0	0	1	120 V Hot Bus	ZipLink Terminal Block (ZL-RTB20-OUT). 1A in-line slow blow fuse	120 V hot feed shared by all outputs
Output 0	OUT0	1	0	0	0	0	A1	TX Contactor (GH15-BN)	Drives TX coil via ZipLink
Output 1	OUT1	1	0	0	0	0	A1	BUS Contactor (GH15-BN)	Drives BUS coil via ZipLink
Output 2	OUT2	1	0	0	0	0	A1	F1 Contactor (GH15-BN)	Drives F1 coil via ZipLink
Output 3	OUT3	1	0	0	0	0	95	Thermal Overload Relay (RTD32-180)	Feeds NC contact 95 $\rightarrow$ 96 $\rightarrow$ F2 coil
Module Power Feed	Backplane	0	0	0	0	1	Slot 3 Connector	PLC base backplane (120 VAC)	No external wiring

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
	Total Points								

Sheet: HMI (EA1-T4CL)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
HMI Serial Comm Port	COM Port	0	0	0	0	0	Serial Port	H2-DM1E	Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits
HMI Power +V	+VDC	0	0	0	0	1	+V	Mean Well NDR-480-24	HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail
HMI Power 0 V	0VDC	0	0	0	0	1	0V	Mean Well NDR-480-24	0 VDC return to Mean Well NDR-480-24 -V rail (system DC common)
HMI Ground	Ground	0	0	0	0	1	Ground	Ground	
	Total Points								

Sheet: Output Module (F2-02DAS-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Analog Common (CH1)	CH1-V	0	0	1	0	0	CM	VFD (GS1-20P2) through Ziplink ZL-RTB20	0V1 and CH1-V are internally shorted
Analog Output CH1	CH1+V	0	0	1	0	0	AL	VFD (GS1-20P2) though Ziplink ZL-RTB20	0–10 V speed command to VFD

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Module Logic +24 V	+V1	0	0	0	0	1	VDC +24V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Powers analog output circuitry, also fused with 1A fast blow fuse
Module Logic 0 V	0V1	0	0	0	0	1	VDC – 0V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Common return for module logic
Slot Power Feed	Backplane	0	0	0	0	1	PLC Base Backplane	Internal slot power	No external wiring
	Total Points								

Sheet: Input Module (D2-08ND3)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X0	X0	0	1	0	0	0	Forward Pushbutton	Forward button S2PR-P3G with Contact block SA-CA with LED module SA-LDG	Shared button +24V -> NO contact -> X0.
D2-08ND3 X1	X1	0	1	0	0	0	Reverse Pushbutton	Reverse button S2PR-P3Y with Contact block SA-CA with LED module SA-LDY	Shared button +24V -> NO contact -> X1.
D2-08ND3 X2	X2	0	1	0	0	0	Run 1 / Deadman 1 Pushbutton	Run 1 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X2.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X3	X3	0	1	0	0	0	Run 2 / Deadman 2 Pushbutton	Run 2 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X3.
D2-08ND3 X4	X4	0	1	0	0	0	E-Stop Button Monitor	E-stop button E22B1 with NC contact block, wired in series with Mitsubishi SD-N35 contactor coil	PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V.
D2-08ND3 X5	X5	0	1	0	0	0	VFD Fault Relay	VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC/FB), programmed as fault indication via PD052=02	PSU +V -> VFD FA. VFD FC -> X5. FB unused.
D2-08ND3 X6	X6	0	1	0	0	0	Reserved - Reset Pushbutton (planned)	Reset button S2PR-P3R with Contact block SA-CA with LED module SA-LDR	Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).
D2-08ND3 X7	X7	0	1	0	0	0	Spare	Unused	Unwired.
D2-08ND3 Input Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Both top-row and bottom-row C terminals on the D2-08ND3 are internally tied.	C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A fast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the common +24V to X0/X1/X2/X3 (and X6 when wired).
Total Points									

Sheet: Analog Input (F2-08AD-1)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08AD-1 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08AD-1 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Shared with VFD DCM and ACM (all tied to PSU -V).	PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.
F2-08AD-1 CH1+ (Channel 1 Signal)	CH1+	0	0	0	1	0	VFD Motor Current Monitor	VFD analog output Huanyang HY02D211B-T VO terminal, configured for output current via PD054=1 (0-10V proportional to motor current)	VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V).
F2-08AD-1 CH2	CH2	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH3	CH3	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH4	CH4	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH5	CH5	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH6	CH6	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH7	CH7	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH8	CH8	0	0	0	1	0	Spare	Unused channel	Unwired.
Total Points									

Sheet: RTD Input (F2-04RTD)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-04RTD +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal. Module installed in rack but no channels currently used in this build.
F2-04RTD 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	PSU -V -> Module 0V terminal.
F2-04RTD CH1	CH1	0	0	0	1	0	Spare	Unused RTD input channel	Unwired. Available for future RTD temperature sensor.
F2-04RTD CH2	CH2	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH3	CH3	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH4	CH4	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
Total Points									

Sheet: Relay Output (D2-08TR)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR Y0	Y0	1	0	0	0	0	Forward Indicator LED	LED module SA-LDG (green, 12-24 VDC) integrated in Forward pushbutton S2PR-P3G (assembly also drives X0)	PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0 contact. Relay sinks Y0 to -V when energized, lighting the LED.

Point Description	**Origin**	**DO**	**DI**	**AO**	**AI**	**Pwr**	**Destination Address**	**Destination Description**	**Notes**
D2-08TR Y1	Y1	1	0	0	0	0	Reverse Indicator LED	LED module SA-LDY (yellow, 12-24 VDC) integrated in Reverse pushbutton S2PR-P3Y (assembly also drives X1)	PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1 contact. Relay sinks Y1 to -V when energized.
D2-08TR Y2	Y2	1	0	0	0	0	Run 1 / Deadman 1 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 1 pushbutton S2PR-P3B #1 (assembly also drives X2)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2 contact.
D2-08TR Y3	Y3	1	0	0	0	0	Run 2 / Deadman 2 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 2 pushbutton S2PR-P3B #2 (assembly also drives X3)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3 contact.
D2-08TR Y4	Y4	1	0	0	0	0	Error / Fault Indicator	Wamco model 525 panel-mount LED indicator (28 VDC, 0.6 W)	PSU +V -> Wamco 525 (+). Wamco 525 (-) -> Y4 contact. 28V-rated indicator slightly under-driven at 24V (acceptable, ~21 mA draw).
D2-08TR Y5	Y5	1	0	0	0	0	VFD Fault Reset	Huanyang HY02D211B-T VFD RST terminal (multi-input, sinking/NPN mode)	Y5 -> VFD RST. When Y5 energized, relay closes Y5 to common (PSU -V), pulling RST input to DCM. ~500 ms pulse during PLC unjam routine.
D2-08TR Y6	Y6	1	0	0	0	0	VFD Forward Run	Huanyang HY02D211B-T VFD FOR terminal (multi-input, sinking/NPN mode)	Y6 -> VFD FOR. When Y6 energized, pulls FOR to DCM. Forward shredding.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR Y7	Y7	1	0	0	0	0	VFD Reverse Run	Huanyang HY02D211B-T VFD REV terminal (multi-input, sinking/NPN mode)	Y7 -> VFD REV. When Y7 energized, pulls REV to DCM. Used during PLC unjam routine.
D2-08TR Relay Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	C -> PSU -V. Sinking mode: when any Y0-Y7 closes, that terminal connects to -V.
Total Points									

Sheet: Analog Output (F2-08DA-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08DA-2 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08DA-2 0V (Module Power Return / Analog Common)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Single 0V terminal serves both as power return and analog signal reference for all 8 channels.	PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through the common 0V rail.
F2-08DA-2 +V1 (Channel 1 Signal)	+V1	0	0	1	0	0	VFD Analog Speed Reference	Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10 V), used as frequency/RPM reference. Requires PD002=1.	+V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08DA-2 +V2	+V2	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V3	+V3	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V4	+V4	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V5	+V5	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V6	+V6	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V7	+V7	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V8	+V8	0	0	1	0	0	Spare	Unused channel	Unwired.
Total Points									

Sheet: VFD (Huanyang HY02D211B-T)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD R (AC Line Hot Input)	R	0	0	0	0	1	AC Mains Hot via main disconnect and 30 A breaker	120 VAC line conductor from wall outlet, routed through panel main disconnect and a 30 A C-curve DIN-rail circuit breaker	Wall AC L -> disconnect -> 30 A breaker -> VFD R. Black wire per NFPA 79.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD T (AC Neutral Input via Contactor and Breaker)	T	0	0	0	0	1	AC Mains Neutral via Phoenix Contact TMC 8 3C 15A breaker and Mitsubishi SD-N35 contactor	Neutral leg switched through the safety-disconnect contactor. As-built path: Wall N -> Phoenix Contact TMC 8 3C 15A breaker (P/N 2907618) -> SD-N35 T1 -> contactor pole -> SD-N35 L1 -> VFD T.	AS-BUILT: neutral leg is switched by the contactor instead of the hot leg. Note: NFPA 79 / NEC standard practice is to switch the HOT leg. Review for final report. When E-stop is pressed, SD-N35 coil drops -> contact opens -> VFD T loses neutral -> VFD shuts down.
VFD Ground (PE)	Ground	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds VFD chassis to panel ground bus	VFD ground stud -> panel ground bus. Green or Green/Yellow wire.
VFD U (Motor Phase 1 Output)	U	0	0	0	0	1	Banana Socket (Blue) -> Motor T1	Panel-mount banana socket (blue, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T1	VFD U -> blue banana socket -> motor terminal.
VFD V (Motor Phase 2 Output)	V	0	0	0	0	1	Banana Socket (White) -> Motor T2	Panel-mount banana socket (white, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T2	VFD V -> white banana socket -> motor terminal.
VFD W (Motor Phase 3 Output)	W	0	0	0	0	1	Banana Socket (Red) -> Motor T3	Panel-mount banana socket (red, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T3	VFD W -> red banana socket -> motor terminal.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD P+ (Brake Resistor +)	P+	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (+ terminal)	Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO model APCS-300R30-AD)	VFD P+ -> brake resistor terminal 1.
VFD PR (Brake Resistor Return)	PR	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (return terminal)	Dynamic braking resistor (same component as P+ row)	VFD PR -> brake resistor terminal 2.
VFD UPF	UPF	0	1	0	0	0	Unused	Multi-Output 2 (PD051), open-collector optocoupler	Unwired.
VFD DRV	DRV	0	1	0	0	0	Unused	Multi-Output 1 (PD050), open-collector optocoupler	Unwired.
VFD DCM (Digital Common, top row)	DCM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for FOR/REV/RST sinking inputs	VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.
VFD SPL	SPL	0	1	0	0	0	Unused	Multi-Input 6 - default low-speed select	Unwired.
VFD SPM	SPM	0	1	0	0	0	Unused	Multi-Input 5 - default middle-speed select	Unwired.
VFD SPH	SPH	0	1	0	0	0	Unused	Multi-Input 4 - default high-speed select	Unwired.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD RST (Fault Reset Input)	RST	0	1	0	0	0	D2-08TR Y5	PLC relay output (sinking NPN mode) driving fault reset	PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.
VFD REV (Reverse Run Input)	REV	0	1	0	0	0	D2-08TR Y7	PLC relay output (sinking NPN mode) driving reverse run command	PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.
VFD FOR (Forward Run Input)	FOR	0	1	0	0	0	D2-08TR Y6	PLC relay output (sinking NPN mode) driving forward run command	PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.
VFD ACM (Analog Common, top row)	ACM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for analog signals	VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out).
VFD VO (Analog Output, 0-10 V)	VO	0	0	1	0	0	F2-08AD-1 CH1+	PLC analog input module CH1, used as motor current monitor (PD054=1)	VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.
VFD 10V	10V	0	0	1	0	0	Unused	+10 V reference output (for external potentiometer)	Unwired.
VFD FA (Multi-Output 3 Relay NO)	FA	0	1	0	0	0	Mean Well NDR-480-24 +V	DC power supply +V rail	PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02).
VFD FC (Multi-Output 3 Relay Common)	FC	0	1	0	0	0	D2-08ND3 X5	PLC discrete input X5 (jam signal)	VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).
VFD FB (Multi-Output 3 Relay NC)	FB	0	1	0	0	0	Unused	Normally-closed contact - not used in active-high NO configuration	Unwired.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD 24V (Onboard 24V Supply)	24V	0	0	0	0	1	Unused	Onboard 24 VDC auxiliary supply, max 200 mA	Unwired. Panel uses external Mean Well NDR-480-24 instead.
VFD DCM (bottom row)	DCM (bottom)	0	0	0	0	1	(same node as top-row DCM)	Internally tied to top-row DCM	Alternate landing point. Top-row DCM is used.
VFD 5V	5V	0	0	1	0	0	Unused	+5 V reference output	Unwired.
VFD ACM (bottom row)	ACM (bottom)	0	0	0	0	1	(same node as top-row ACM)	Internally tied to top-row ACM	Alternate landing point. Top-row ACM is used.
VFD AI (Analog Current Input, 4-20 mA)	AI	0	0	0	1	0	Unused	Analog current input for speed reference. Speed command uses VI voltage input instead.	Unwired.
VFD VI (Analog Voltage Input, 0-10 V)	VI	0	0	0	1	0	F2-08DA-2 +V1	PLC analog output module CH1, used as commanded frequency reference (PD002=1)	F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.
VFD RS-	RS-	0	0	0	0	1	Unused	RS-485 Modbus communication negative line	Unwired. Modbus not used.
VFD RS+	RS+	0	0	0	0	1	Unused	RS-485 Modbus communication positive line	Unwired.
Total Points									

16.3 Appendix C: PLC Ladder Logic — Reference

The complete 31-rung Do-more Designer ladder export is reproduced inline in Section 6.1 (PLC Ladder Logic Architecture), with each page of `electrical/Circuit_logic.pdf` placed adjacent to the rung-by-rung walkthrough that describes its contents. The full PDF source is also retained at `electrical/Circuit_logic.pdf` in the repository for archival and detailed inspection.

16.4 Appendix D: Wiring Diagrams

[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]

16.5 Appendix E: VFD Parameter Settings

The full programmed parameter set — code, function, value, and rationale — is given in Table 1 (Section 5.2, Motor and VFD). Motor values PD141–144 reflect the GE 5KE182BC205B test motor; the production motor is TBD and its nameplate will supersede these.

Power / braking wiring: R = AC hot (30 A breaker); T = AC neutral via SD-N35 contactor; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30 Ω brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

[NOTE: Firmware-specific finding: `PD052 = 12` (continue-running over-torque indication) did not fire the relay on our Huanyang revision; the working configuration is `PD052 = 02` with `PD123 = 3` (over-torque self-trips, which fires the fault relay and is read by the PLC on X5).]

16.6 Appendix E: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

ID	Test case
Top-down	Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY
BU-01	AC power input and fuse distribution
BU-02	24 VDC PSU output (Mean Well NDR-480-24)
BU-03	PLC rack initialization (DirectLOGIC 205)
BU-04	E-stop hardware safety circuit
BU-05	Two-hand start (deadman) safety logic
BU-06	Lid interlock (n/a in this build — access guarded mechanically by the ME team)
BU-07	VFD speed command and motor RPM (HY02D211B-T)
BU-08	Overcurrent / overload protection
BU-09	HMI operator interface (EA1-T4CL C-More Micro-Graphic)
BU-10	Status indicator lights (power / running / fault)

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

17 Authors and Acknowledgements

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Disclaimer

This report documents an in-progress senior capstone design. Findings, parameters, and as-built details reflect the current state of the prototype and are subject to revision through final validation.

End of rough draft. Comments and questions welcome.
